



BITS Pilani

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Product Design

(AE ZG531 / AEL ZG531 / DE ZG541 / DM ZG541 / MM ZG541)



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Lecture-1: Overview of Handout; Introduction to Product Design and Development



Course Information

Course Details



Course: Product Design

Code: AE ZG531, AEL ZG531, DE ZG541, DM ZG541, MM ZG541

Units: 5

Instructor-in-charge:

Samata Satish Mujumdar – Associate Professor Core Engineering

samata.mujumdar@pilani.bits-pilani.ac.in

Vijay Kumar B R B

vijaykumar.brb@wilp.bits-pilani.ac.in

Course Description



- Introduction to Product Development (PD)
- Product Development Planning
- Customer Requirements
- Develop Product Specifications
- Product Portfolios & Product Architecture
- Concept Design Study
- Detailed Design Study
- Industrial Design
- Design for Material, Manufacturing and Assembly
- Design for Environment
- Design for Safety, Reliability and Maintainability
- Product Development Economics
- CAE and PLM

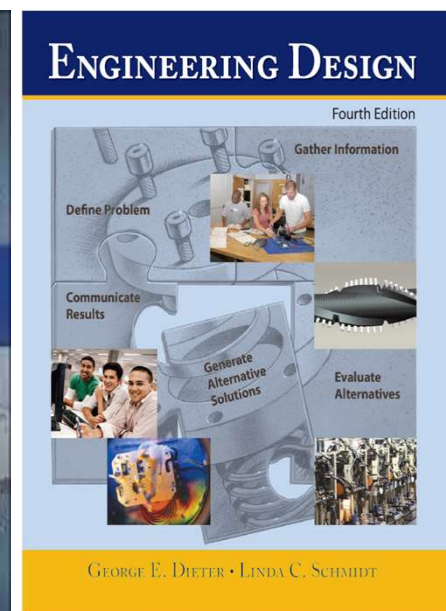
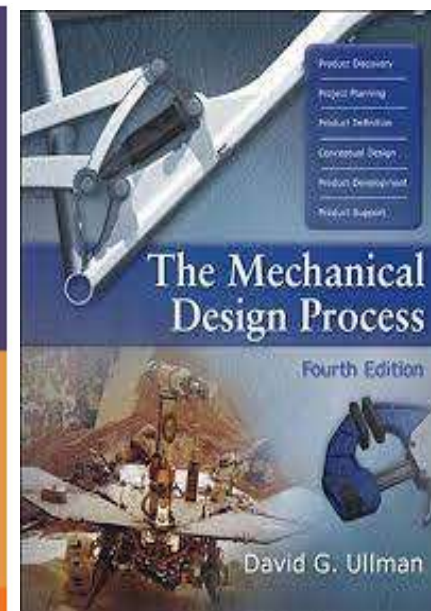
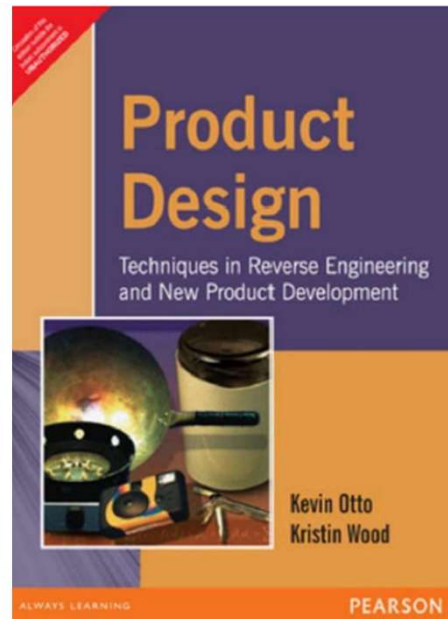
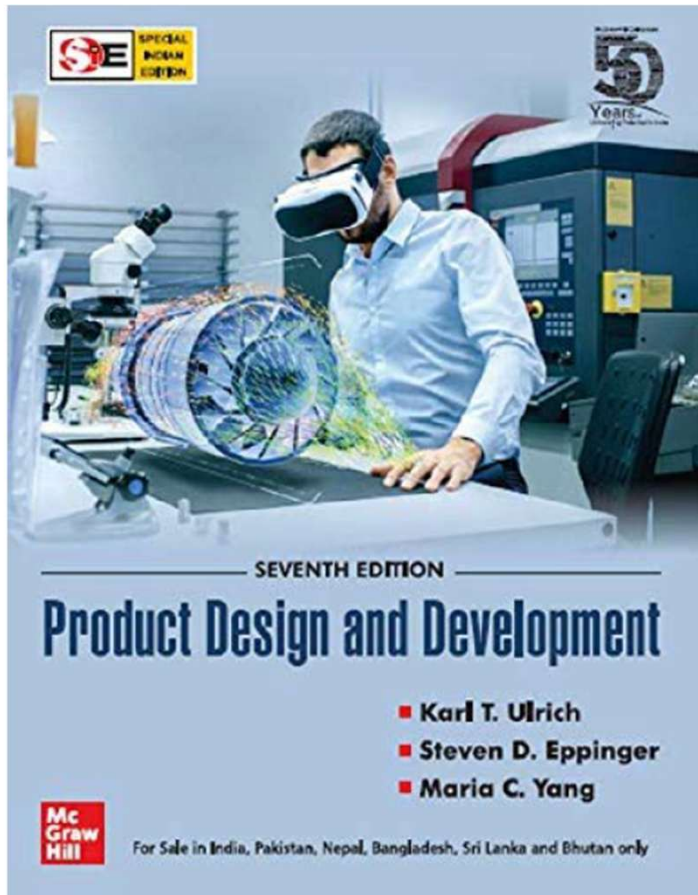
Course Objectives



Students will be able to, but not limited to:

- (1) Understand entire PD cycle
- (2) Integrate knowledge from marketing, design, and manufacturing to develop product strategies.
- (3) Understand and work towards 'Design for X'
- (4) Relate industrial design, prototyping techniques, and market analysis
- (5) Manage PD projects

Text Book and Reference Books



Evaluation Scheme



Evaluation Component	Evaluation Name	Type	Marks	Weight (%)	Duration	Date
EC - 1	Theory - Quiz 1 & Quiz 2	Online	30 (15x2)	10%		Quiz1 – Pre Midsem Quiz2 – Post Midsem
	Lab Assignment	Online	45	15%		
	Lab Quiz	Online	15	05%		
EC - 2	Mid-Semester Exam	Closed Book	90	30%	2 hours	
EC - 3	Comprehensive Exam	Open Book	120	40%	2½ hours	

Syllabus:

- Mid-Semester Exam: Session 1 to 8
- Comprehensive Exam: All topics

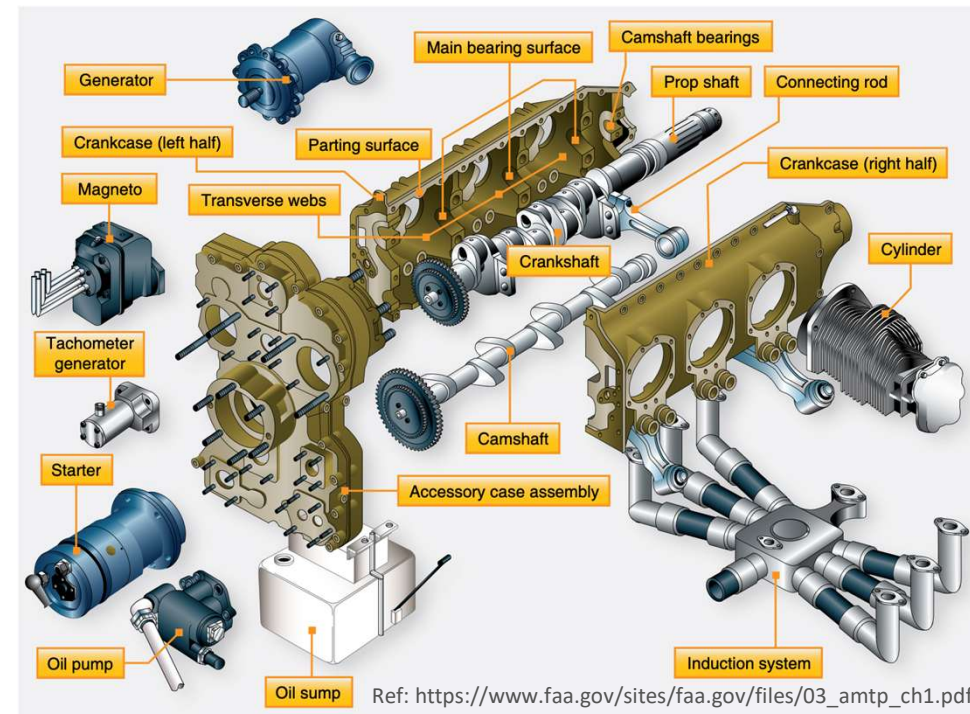


**What is a Product and
Why “Product Development”
is so important?**

Product



- A *Product* is something sold by an enterprise to its customers.
- *Product Development* is the set of activities beginning with the perception of a market opportunity and ending in the production, sale, and delivery of a product



Attributes of Products



	Stanley Tools Jobmaster Screwdriver	Rollerblade In-Line Skate	Hewlett-Packard DeskJet Printer	Volkswagen New Beetle Automobile	Boeing 777 Airplane
Annual production volume	100,000 units/year	100,000 units/year	4 million units/year	100,000 units/year	50 units/year
Sales lifetime	40 years	3 years	2 years	6 years	30 years
Sales price	\$5	\$150	\$130	\$20,000	\$260 million
Number of unique parts (part numbers)	3 parts	35 parts	200 parts	10,000 parts	130,000 parts
Development time	1 year	2 years	1.5 years	3.5 years	4.5 years
Internal development team (peak size)	3 people	5 people	100 people	800 people	6,800 people
External development team (peak size)	3 people	10 people	75 people	800 people	10,000 people
Development cost	\$150,000	\$750,000	\$50 million	\$400 million	\$3 billion
Production investment	\$150,000	\$1 million	\$25 million	\$500 million	\$3 billion

EXHIBIT 1-3 Attributes of five products and their associated development efforts. All figures are approximate, based on publicly available information and company sources.

Products are Always Evolving



OUT-OF-PRODUCTION MODELS



Wood and linen aircraft



All-metal aircraft



All-metal or metal with some composite

AIRCRAFT IN PRODUCTION



All-metal, metal with some composite or mostly composite aircraft

Ref: The Boeing Company; "Boeing Aircraft Since 1916" by Peter M. Bowers, Aero Publishers



100 Years: The evolution of automobile design.



1900's



1920's



1940's



1960's



1980's



2000's



...

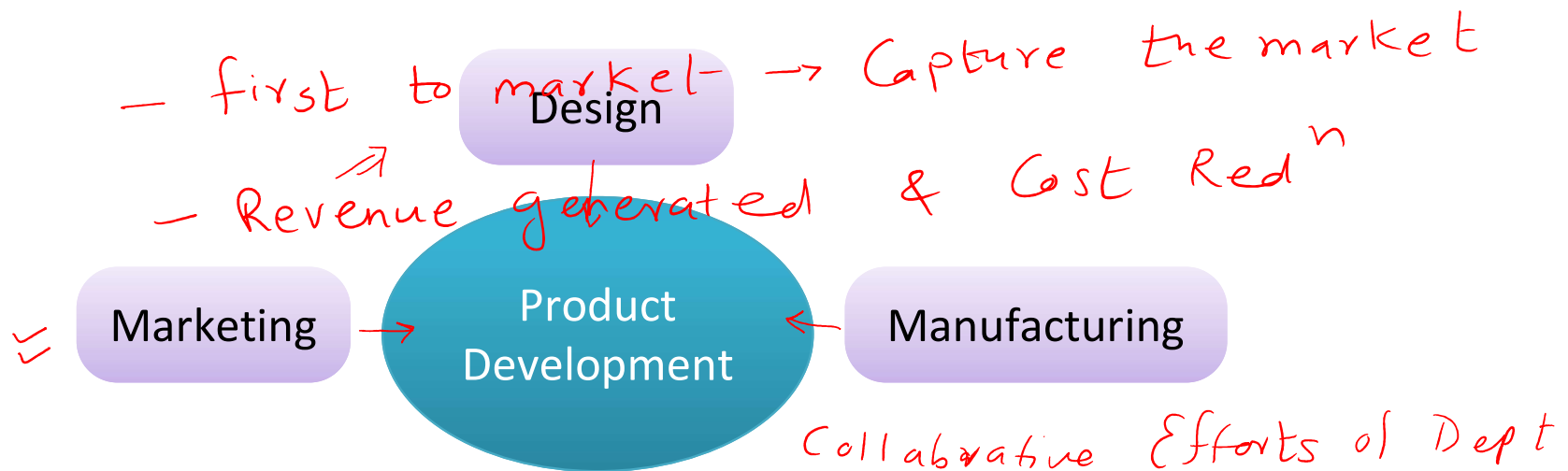
Economic Success of Firms

⇒ Generate money

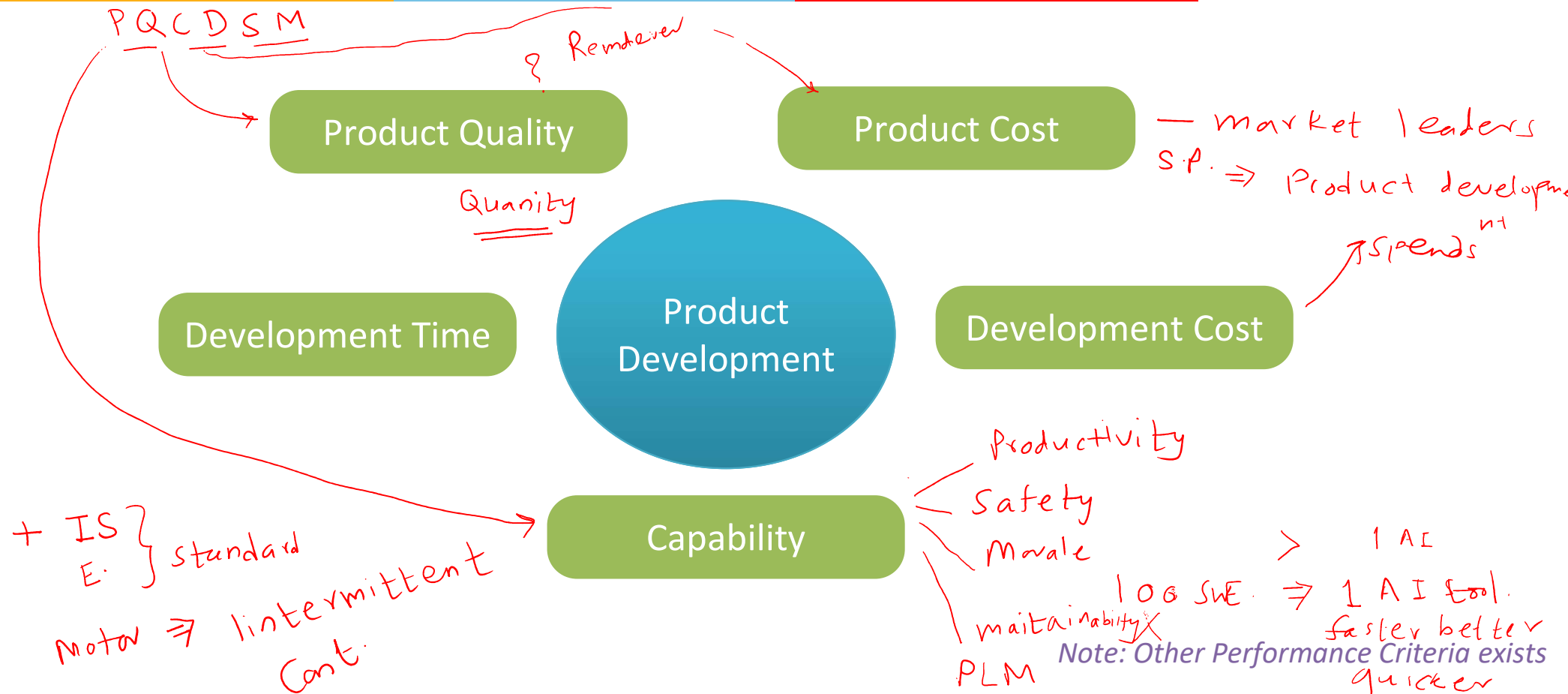


The economic success of most firms depends on:

- (1) Ability to identify the need of customers
- (2) Ability to launch the product quickly that meets customer requirements
- (3) Ability to make and sell the product at lower cost

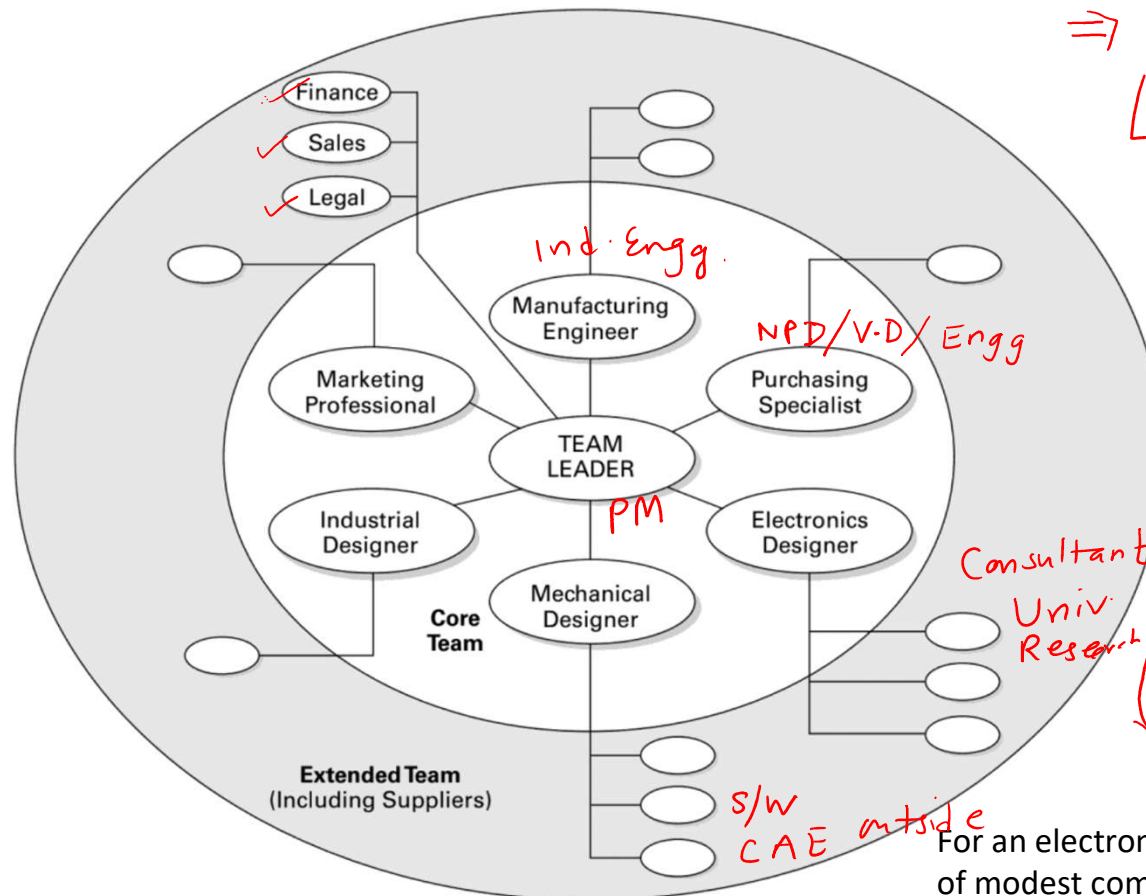


Characteristic of Product Development



Example of a PD Team

IT → Projects based
Automotive / Aerospace / Energy sector



⇒ functional.



project based
matrix

Consultant
Univ.
Research

Accountability
with fⁿ head
Authority with
P.M.

For an electromechanical product
of modest complexity

Challenges of Product Development



Regulation $\Rightarrow$ ^{sp. logistic $\Rightarrow$ lengthy} specific material $\Rightarrow$ costly
" mfg.

Local / Global

B/O vs
Inhouse Development Vs Q.D.
time-long
material Vs Cost
logistic Vs lengthy
type

Trade-offs

Dynamics

Need $\rightarrow$ idea $\rightarrow$ final product

Details

Product Development

Time Pressure

trade off
optimum

product $\swarrow$ multiple
use
Right product at Right time

Economics

Quicker better faster

Revenues / profit

Wanderella

June $\rightarrow$ Oct
Launch

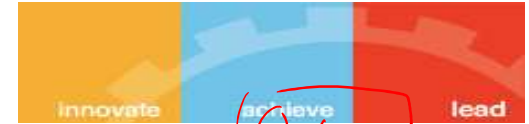
Lost market
man com $\Rightarrow$ Summer product

Course Summary

ISRO, NASA

5 phase of P.D

Need → f^n → technical sp. → D_{imp} / $Lang$
 overall $V \& V$



Ovg

Vision

↓ ↑

Mission

Long term goals

↓

Short terms

↓

Projects

P.D. / P.U. / NPD
 Existing R&D
 Cost & any

Technology led org.

Scientific Principles
 R&D

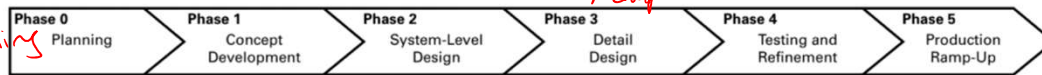
Technology

products

Risk

D time ↑↑
 success not guaranteed.

preplanning



Chapter 2: Development Processes and Organizations

Chapter 3: Opportunity Identification

Chapter 4: Product Planning

Chapter 5: Identifying Customer Needs

Chapter 6: Product Specifications

Chapter 7: Concept Generation

Chapter 8: Concept Selection

Chapter 9: Concept Testing

Chapter 10: Product Architecture

Chapter 11: Industrial Design

Chapter 12: Design for Environment

Chapter 13: Design for Manufacturing

Chapter 14: Prototyping

Chapter 15: Robust Design

Chapter 16: Patents and Intellectual Property

Chapter 17: Product Development Economics

Chapter 18: Managing Projects

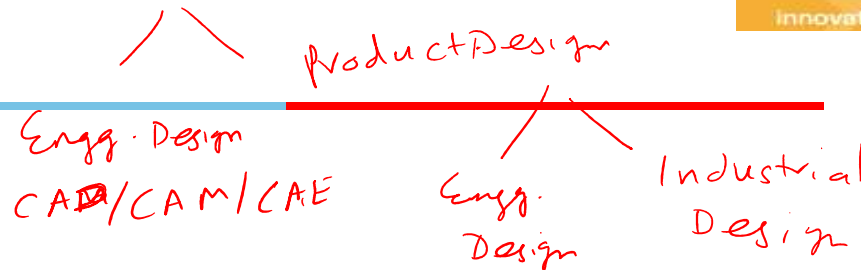
Many More-Focused Development Methods

PPAP & PAF



- **Introduction**
- **Type of Design**
- **Consideration of Good Design**

Introduction



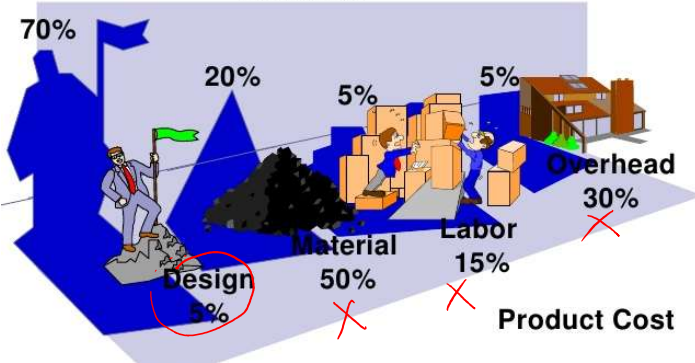
- What is design?
 - To design is to pull together (synthesis) something new or to arrange existing things in a new way.
 - Design establishes and defines solutions to and pertinent structures for problems not solved before, or new solutions to problems which have previously been solved in a different way
 - Good design requires analysis and synthesis
 - Four 'C' of design: Creativity, Complexity, Choice and Compromise

Impact of Engineering Design Process

Product Cost

Who Casts the Biggest Shadow?

Influence

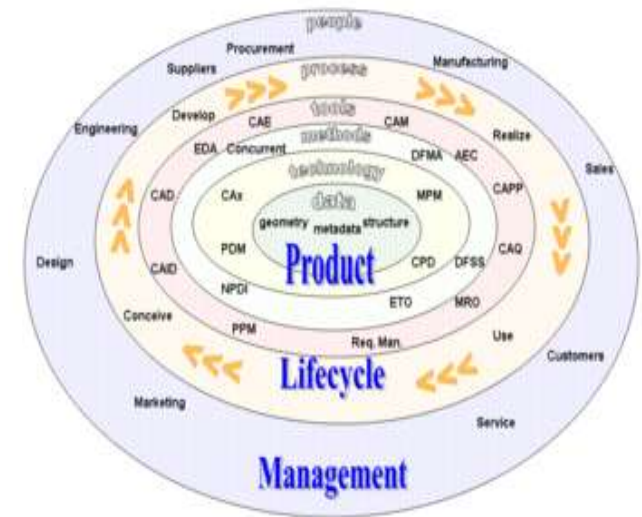


Product Quality

Influence of QbD on phases of the product lifecycle⁴



Product Cycle Time



Types of Design



- (1) Original Design (Innovative design)
- (2) Adaptive Design
- (3) Re-design (Variant Design)
- (4) Selection Design
- (5) Industrial Design

Original Design



- Original design generates original concept for solving specific need, and a few successful original designs occur over time.
- They can disrupt the market, by replacing old equipment, manufacturing methods etc.

Example:

Refrigeration, Air conditioning industry used to work on harmful Ozone depleting Chloro Fluoro Carbon gases (CFCs) such as R-12, R-11 etc and HCFCs such as R-22, R-502 Development of Zero ozone depleting HFCs (Hydro Fluoro carbons) such as R-410A, R-134a impacted the industry to change majority of components, processes etc



Reduced

*sugar management
wearable*

Stay informed with high & low heart rate notifications.



Fitbit, revolutionised the industry of wearable health tech

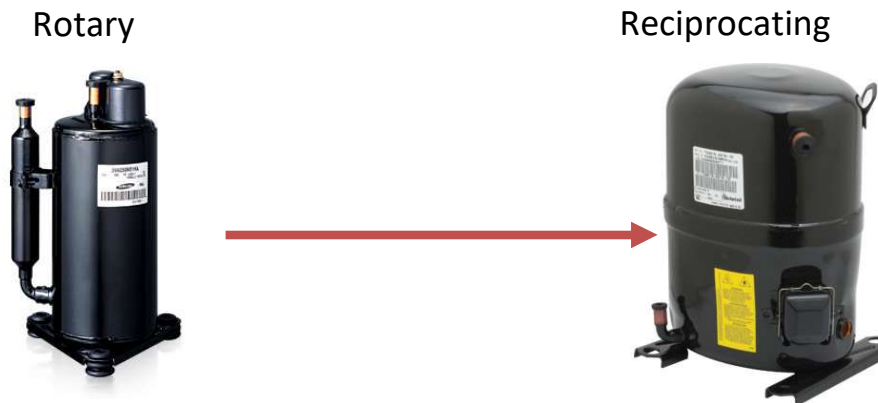
Adaptive Design



Adapting an existing system to a new problem

Example:

Domestic air conditioning market used to be dominated by reciprocating compressors (for compressing refrigerant). Development of Rotary compressors (Size /weight being only half), that can work in harsh climates, replaced most of reciprocating compressors



Re-Design



- Extension of product platforms or improvement in existing design
- Variation of size, geometry, material properties, control parameters etc

Example:

$100 + 0.5$
 $\hookrightarrow 100 + 0.2$

circular
 $\downarrow$
elliptical.

Alloying
H.T.

- (1) Company is manufacturing motors till 15 kW. Extending the product range to 100 kW is a variant design
- (2) Development of aircraft engine fan case in composite, instead of metal.

Selection Design



Selection of off-the-shelf components with the needed properties

shaft / keys / bearing / screws / spacers / nuts / dowels / pins

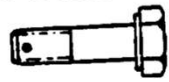
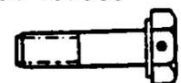
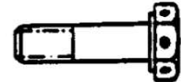
Example:

- (1) Bearings of specific load and speed
- (2) Standard fasteners (bolts and nuts), that are used to connect aircraft engine fan cases

SKF bearing

standards

bolts / spare part ⇒ select ⇒

AN105501-106400 	Bolt, hex. Head, drilled shank, corrosion-resistant steel
AN106401-107300 	Bolt, hex., drilled head, (one holes), corrosion-resistant steel
AN107301-108200 	Bolt, hex., drilled head, (six holes), corrosion-resistant steel
MS9033-9039 	Bolt, machine 12pt. Head, 130,000 psi min. T.S.

Ref: https://www.faa.gov/sites/faa.gov/files/2023-08/AC_43.13-1B_Ch7.pdf

Industrial Design



Product that are designed for the market, that primarily focuses on form, aesthetic, ergonomics for mass usage. *finish*

human factors Engg / Anthropometric data

Example:

- (1) Chairs
- (2) Shoes



Problem Solving Methodology



Challenge of Engineering Design Process

Definition of the problem

Need → Concept → Mfg.

Gathering of Information

*→ Need
Classify. Primary
secondary
tertiary*

Generation of alternative solutions

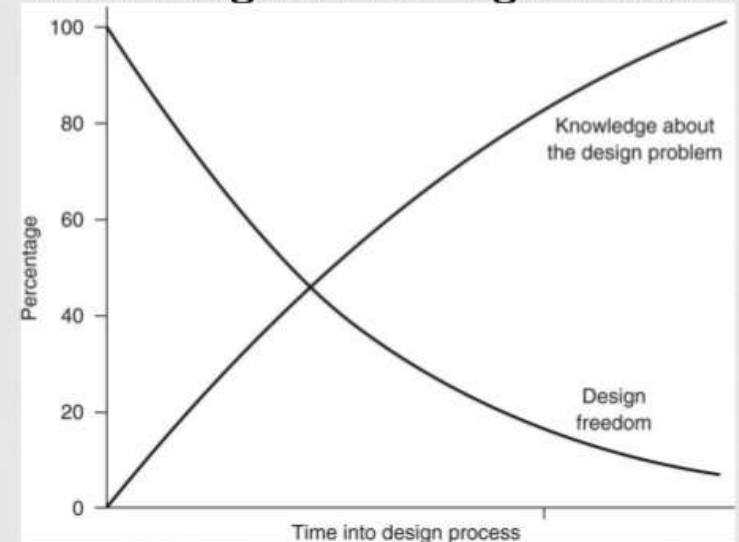
Need → fⁿ & form : solutions → product

Evaluation of alternatives and decision making

PRODUCT

Communication of the results

Design Paradox Between Design Knowledge and Design Freedom



Dieter/Schmidt, Engineering Design Se.
© 2013, The McGraw-Hill Companies

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Considerations of a Good Design



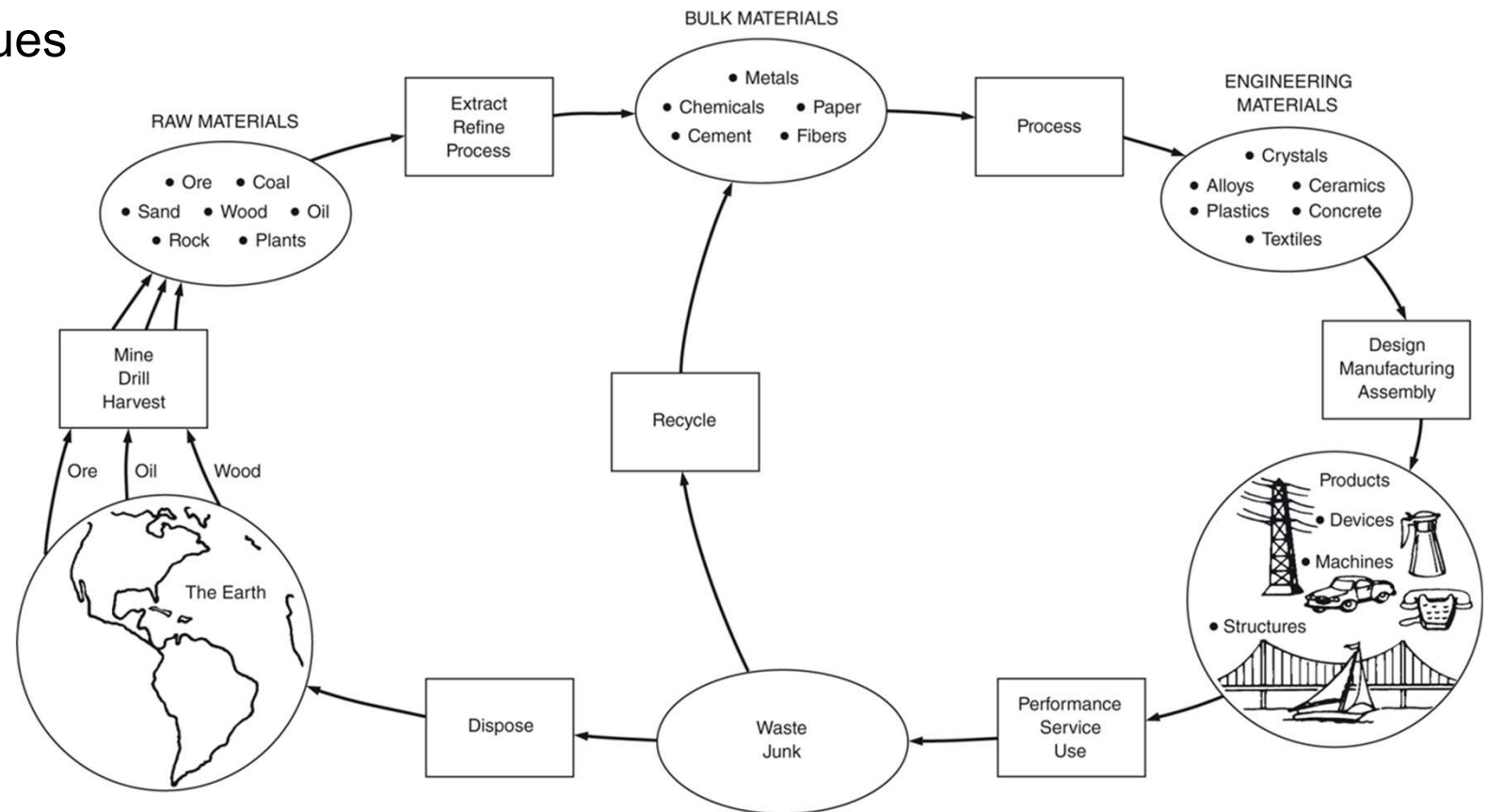
(1) Achievement of performance requirements:

- i. Primary performance requirements
- ii. Complementary performance requirements
 - Useful life of the design, Reliability, Ease, Robustness, Safety and Maintenance.
- iii. Environmental requirements for performance
 - Service conditions under which the product must operate
 - How the product will behave with regard to maintaining a safe and clean environment (Green design, Sustainable design)
- iv. Aesthetic requirements

Considerations of a Good Design



(2) Life-cycle issues



Considerations of a Good Design



(3) Regulatory and Social Issues



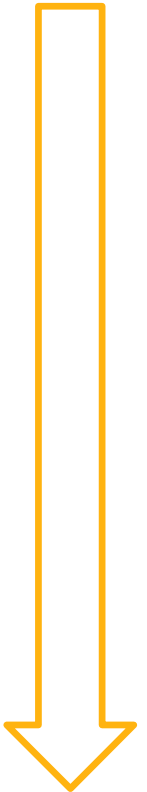
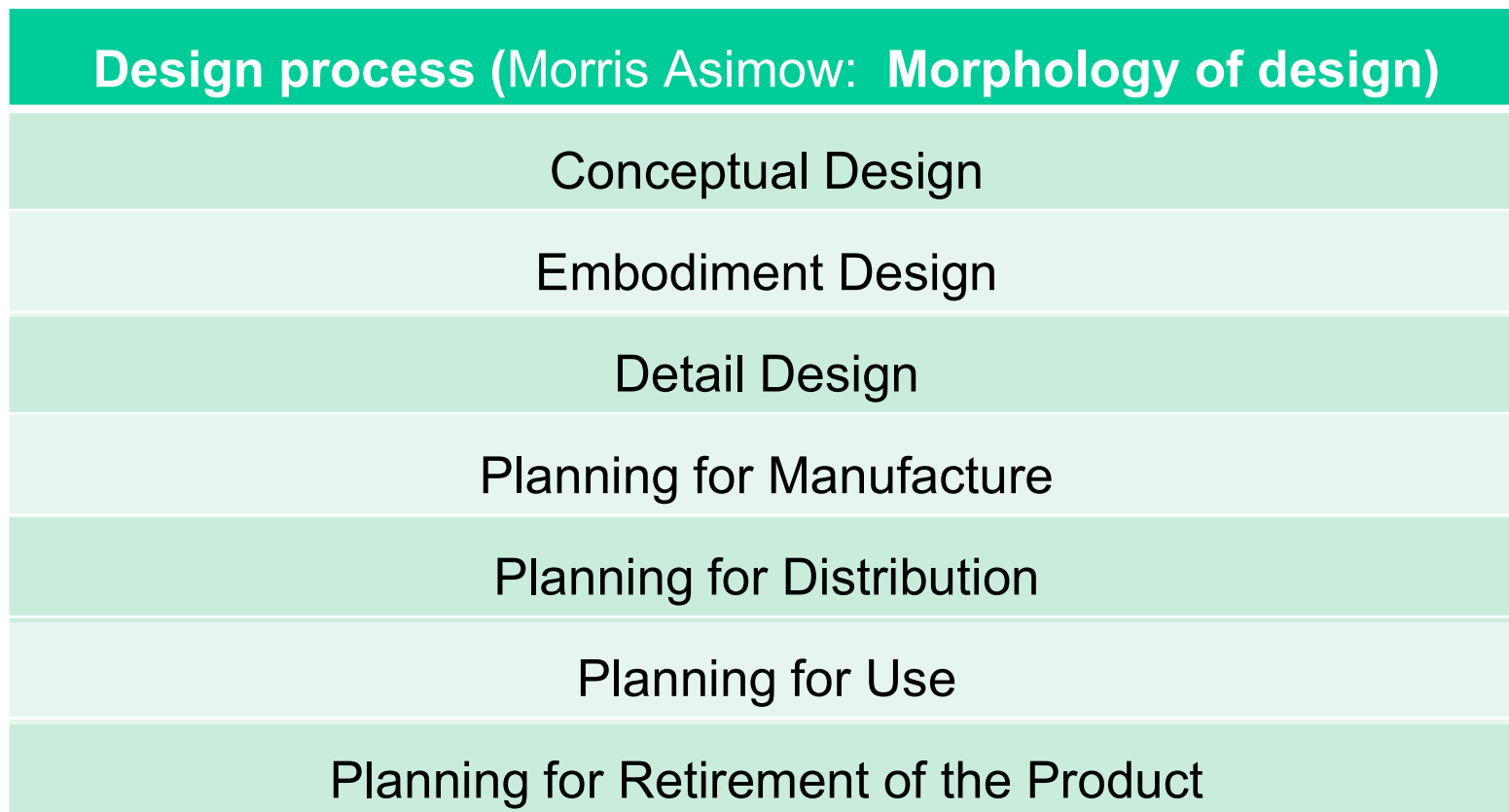
**Federal Aviation
Administration**



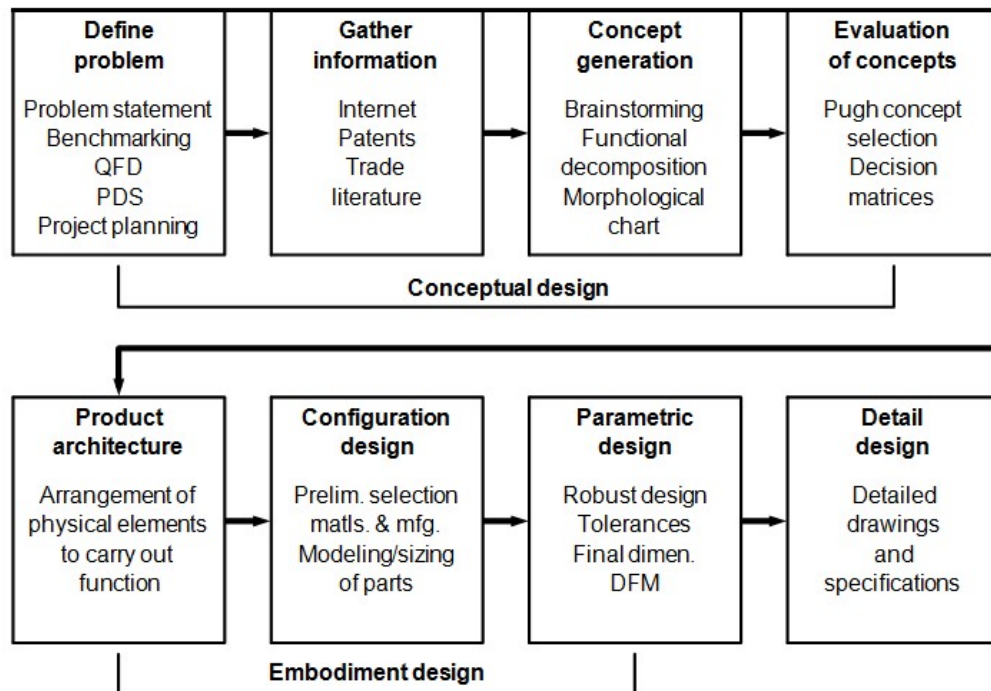


Design Process and PD Process

Description of Design Process



Description of Design Process



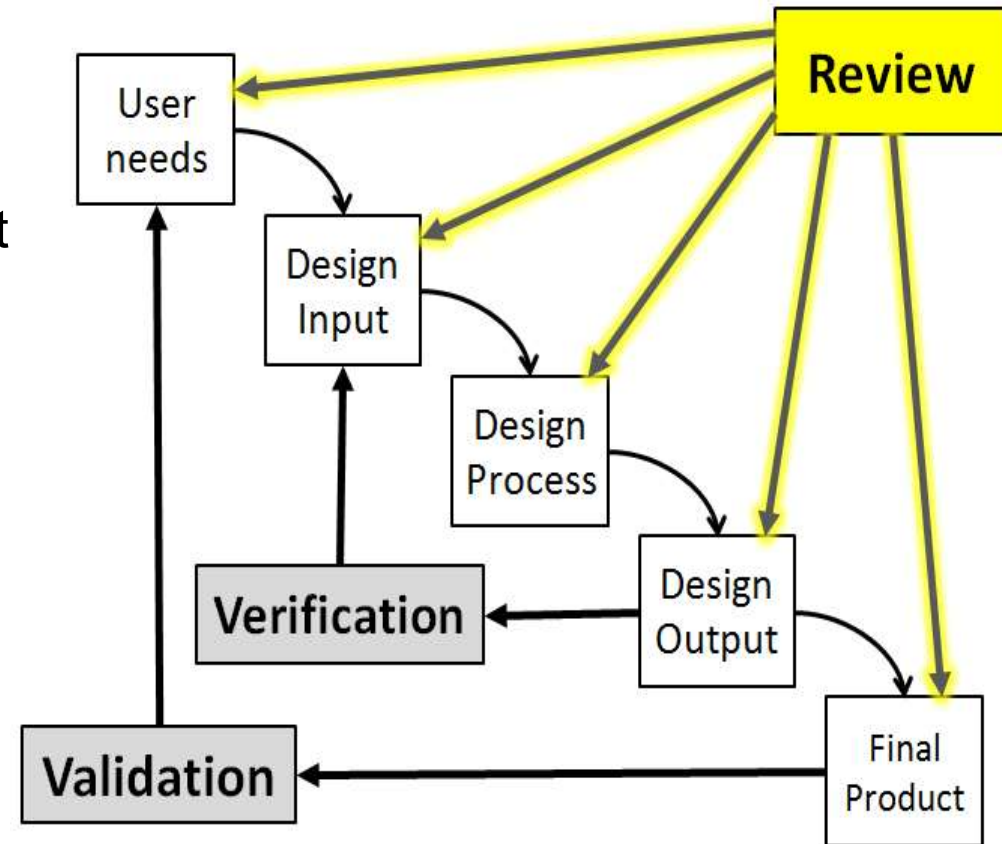
- Phase I: Conceptual Design
- Phase II: Embodiment Design
- Phase III: Detail Design
- Phase IV: Planning for Manufacture
- Phase V: Planning for Distribution
- Phase VI: Planning for Use
- Phase VII: Planning for Retirement of the Product

Design Review



- Retrospective study of the design
- Review team
- Three to six times in the life of the project
- Technical and Business aspects

Are we building it right and building the right product



Product Development Process

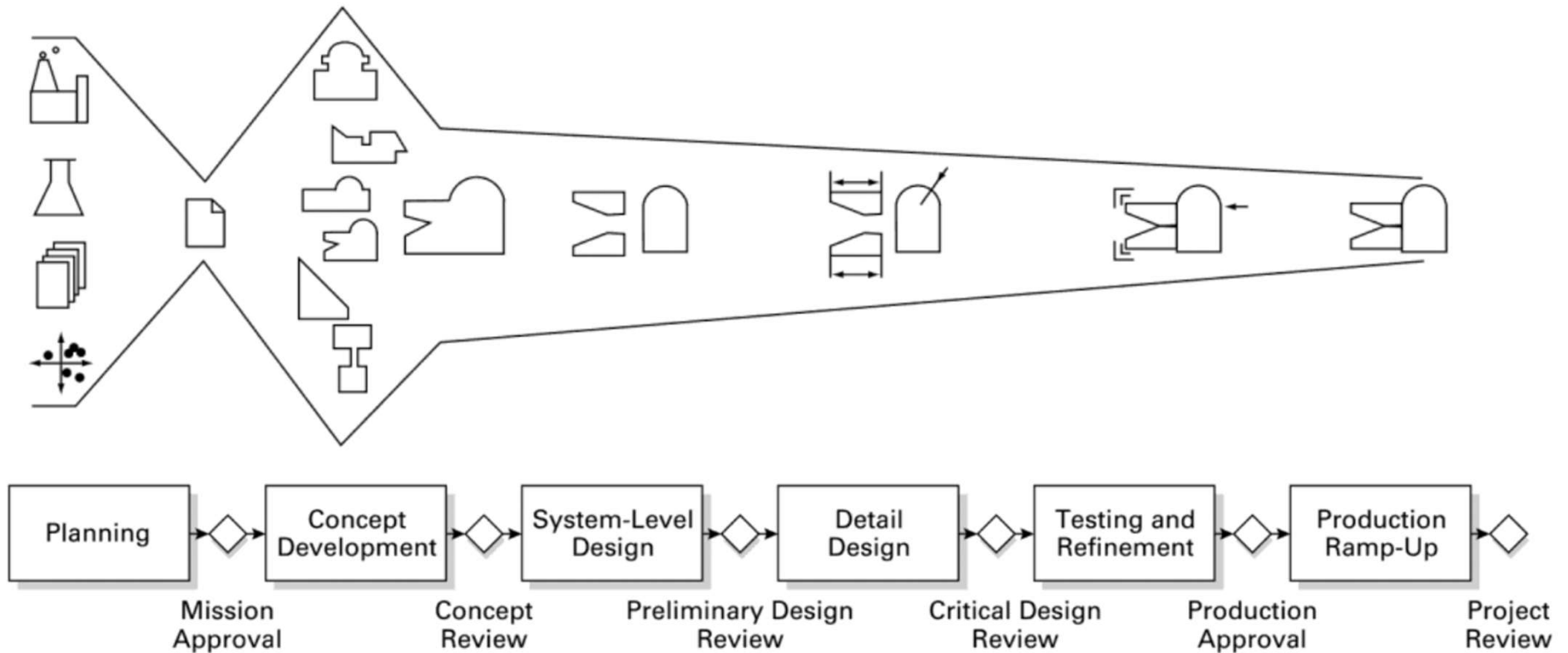


Product Development Process is a sequence of steps or activities that an enterprise employs to conceive, design, and commercialise a product.

Types of PD process:

1. Stage gate development process
2. Spiral product development process
3. Hybrid product development process

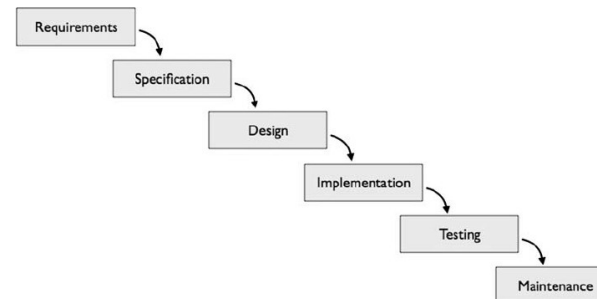
Stage Gate (Generic) PD Process



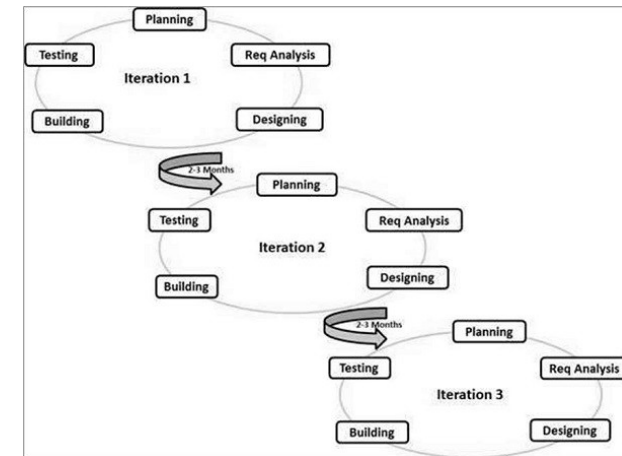
Spiral PD Process



Spiral model = {(1st stage-gate process)+1st version of working product}+{(2nd stage-gate process)+2nd version of working product}+...+{(nth stage-gate process)+nth version of working product}



Waterfall Model



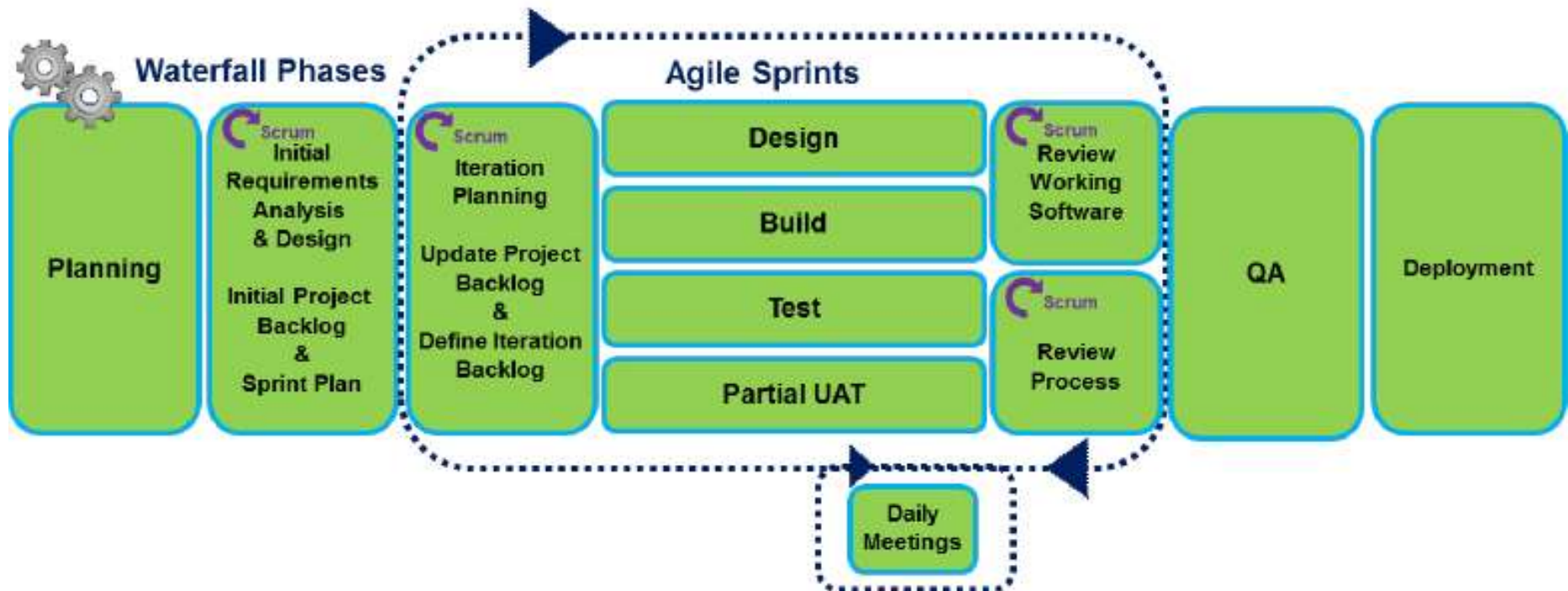
Agile Model

Spiral Model emphasises iterative development and allows for budget adjustments based on feedback and insights gained during each iteration.

Hybrid PD Process



Hybrid Waterfall and Agile Approach



Factors for Success



Success of a product:

- The quality, performance, and price of the product.
- The cost to manufacture the product over its life cycle.
- The cost of product development.
- The time needed to bring the product to the market.

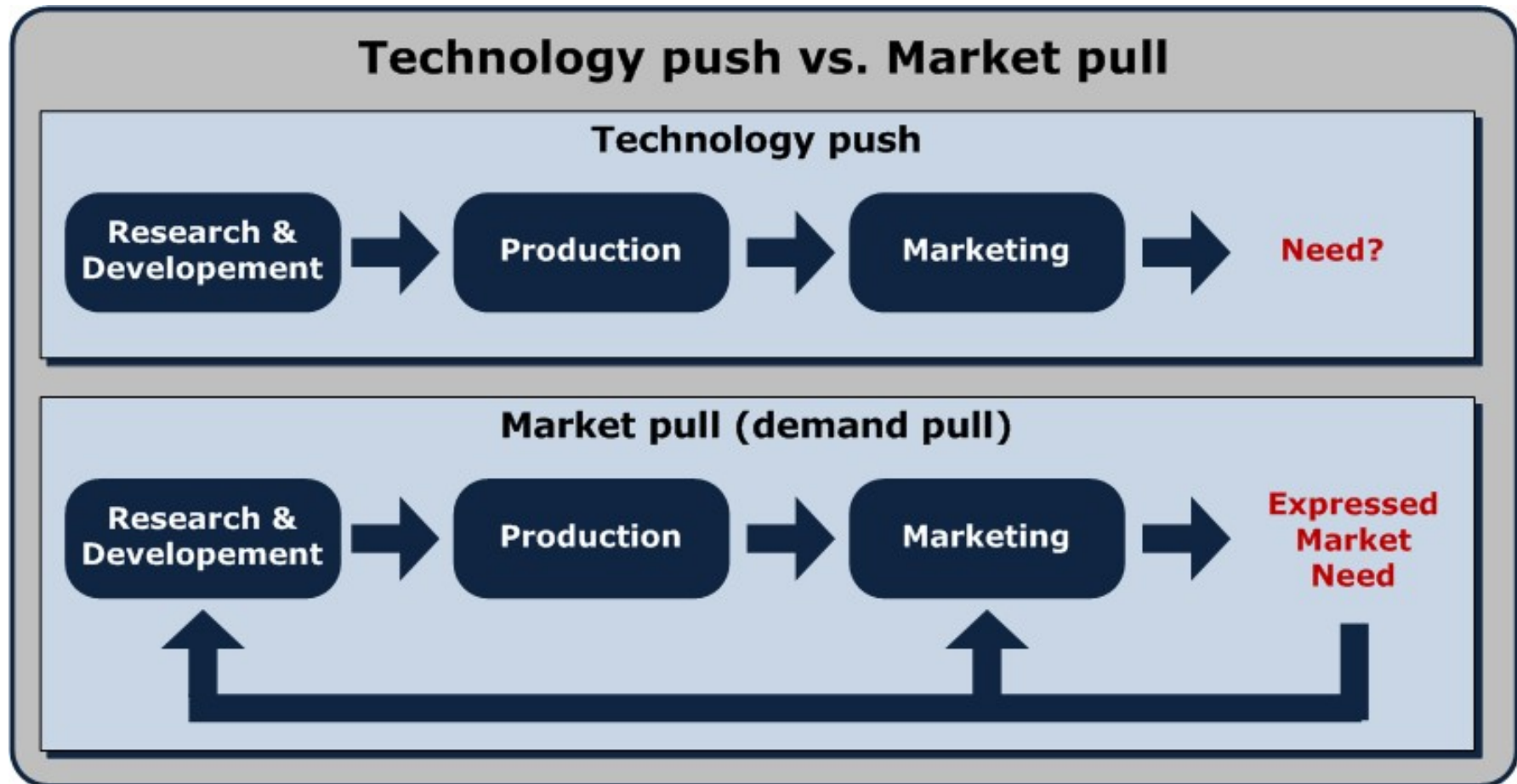


- $\text{Profit} = \text{Product Price} (\uparrow) - \text{Product Cost} (\downarrow)$



Generic PD Process

Variations in Generic PD Process



Adapting the Generic PD Process



Process Type	Description	Distinct Features	Examples
Generic (Market-Pull) Products	The team begins with a market opportunity and selects appropriate technologies to meet customer needs.	Process generally includes distinct planning, concept development, system-level design, detail design, testing and refinement, and production ramp-up phases.	Sporting goods, furniture, tools.
Technology-Push Products	The team begins with a new technology, then finds an appropriate market.	Planning phase involves matching technology and market. Concept development assumes a given technology.	Gore-Tex rainwear, Tyvek envelopes.
Platform Products	The team assumes that the new product will be built around an established technological subsystem.	Concept development assumes a proven technology platform.	Consumer electronics, computers, printers.

Adapting the Generic PD Process



Process Type	Description	Distinct Features	Examples
Process-Intensive Products	Characteristics of the product are highly constrained by the production process.	Either an existing production process must be specified from the start, or both product and process must be developed together from the start.	Snack foods, breakfast cereals, chemicals, semiconductors.
Customized Products	New products are slight variations of existing configurations.	Similarity of projects allows for a streamlined and highly structured development process.	Motors, switches, batteries, containers.
High-Risk Products	Technical or market uncertainties create high risks of failure.	Risks are identified early and tracked throughout the process. Analysis and testing activities take place as early as possible.	Pharmaceuticals, space systems.

Adapting the Generic PD Process



Process Type	Description	Distinct Features	Examples
Quick-Build Products	Rapid modeling and prototyping enables many design-build-test cycles.	Detail design and testing phases are repeated a number of times until the product is completed or time/budget runs out.	Software, cellular phones.
Complex Systems	System must be decomposed into several subsystems and many components.	Subsystems and components are developed by many teams working in parallel, followed by system integration and validation.	Airplanes, jet engines, automobiles.

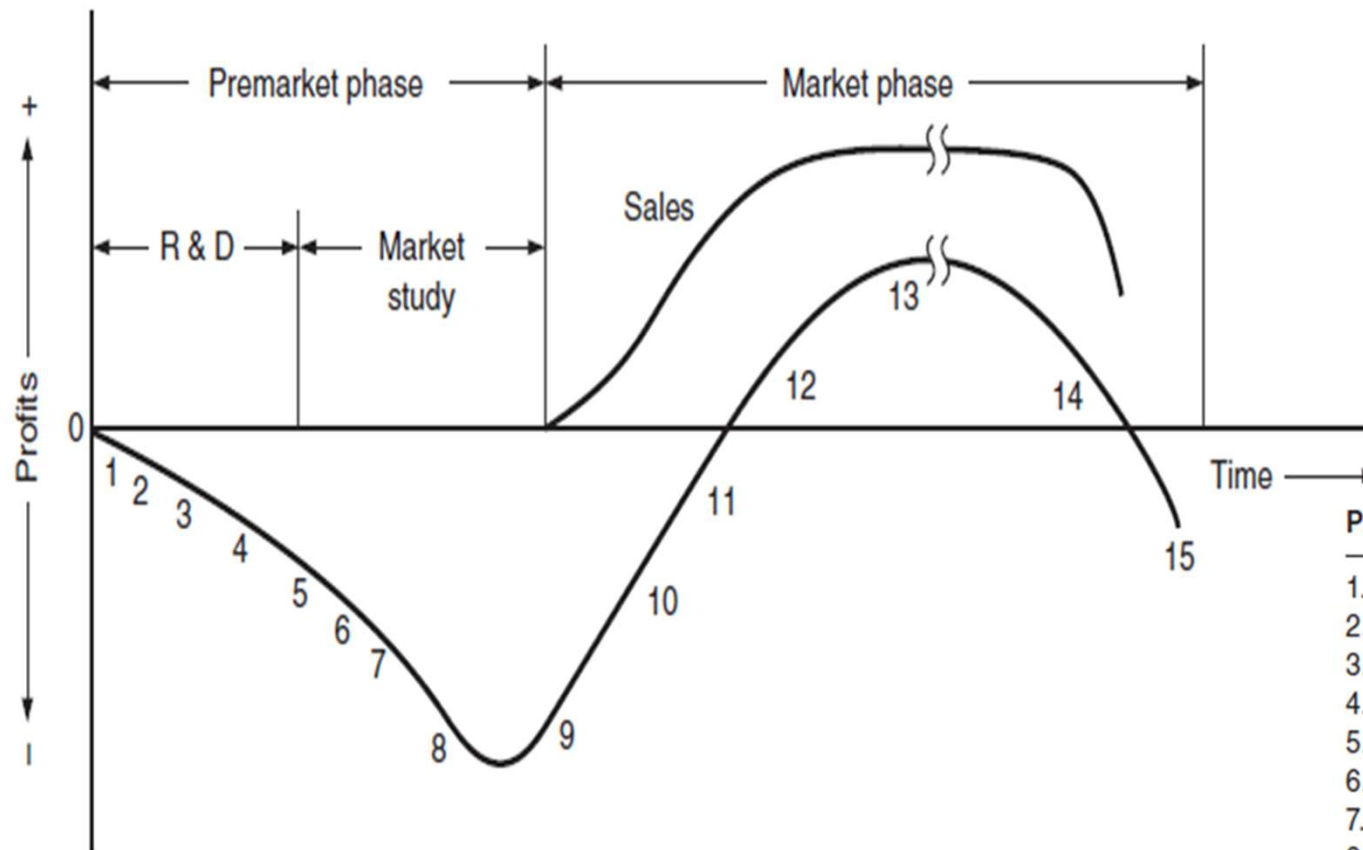
Technology Push Vs. Market Pull



Activity:	Technology Push:	Market Pull:
Business Decisions	Risky	Safe
Cost of Risk	Moderate to High	Low to Moderate
Cost of R&D	Moderate to High	Low to Moderate
Return on Investment (ROI)	Unknown, unsure	Assured
Cost of Marketing	Reshaping the Market	Easy
Investment	Moderate to High	Low to Moderate
Market Education	Customer educated by producer	Customer educated by customers
Managerial Mindset	Vision	Leadership



Technology Development and Insertion



Premarket phase

1. Idea generation
2. Idea evaluation
3. Feasibility analysis
4. Technical R&D
5. Product (market) R&D
6. Preliminary production
7. Market testing
8. Commercial production

Market phase

9. Product introduction
10. Market development
11. Rapid growth
12. Competitive market
13. Maturity
14. Decline
15. Abandonment



Summary

Summary



Phases in Engineering Design Process



Summary



- The keys to creating a winning product are:
 - Designing a quality product with the features and performance desired by its customers at a price they are willing to pay
 - Reducing the cost to manufacture the product over its life cycle
 - Minimizing the cost of developing the product
 - Quickly bringing the product to market
 - The organization of a product development team can have a major influence on how effectively product development is carried out.
 - Generally, a heavyweight matrix organization with appropriate management controls works best

Summary



- Every product development process is different. Its depends on companies technological and market environment.
- Engineers need to develop their own product development process in any business they work and must continually strive to improve it.
- Reverse engineering and redesign is a forum for learning, experimenting and living product design process.
- Handful of reverse engineering experience make you to understand how product are executed within their “Black boxes”.



Thank You!



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Lecture-2: Product Development Process and Planning



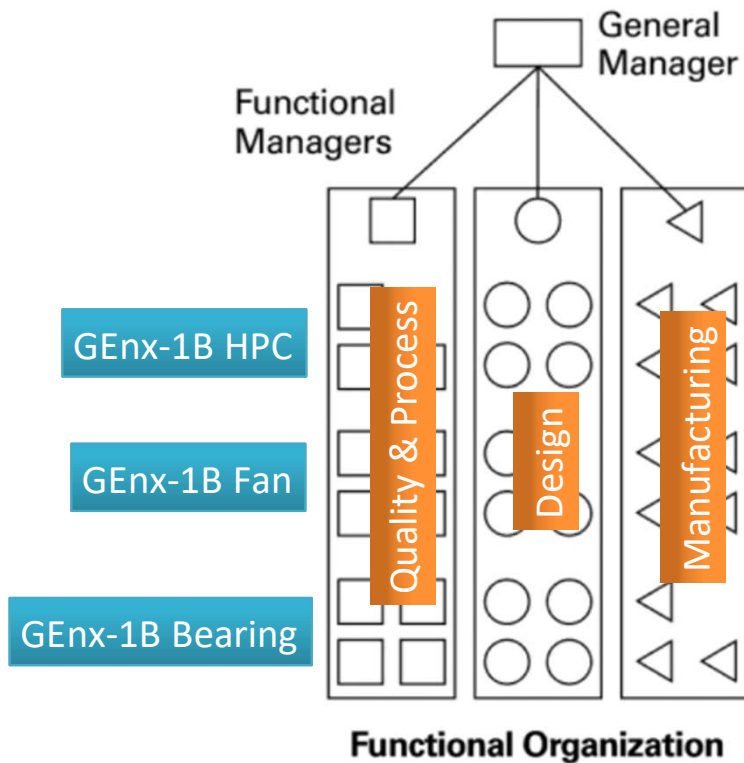
Topics to be covered:

- Product Development Organization
- Concurrent Engineering
- Product Planning

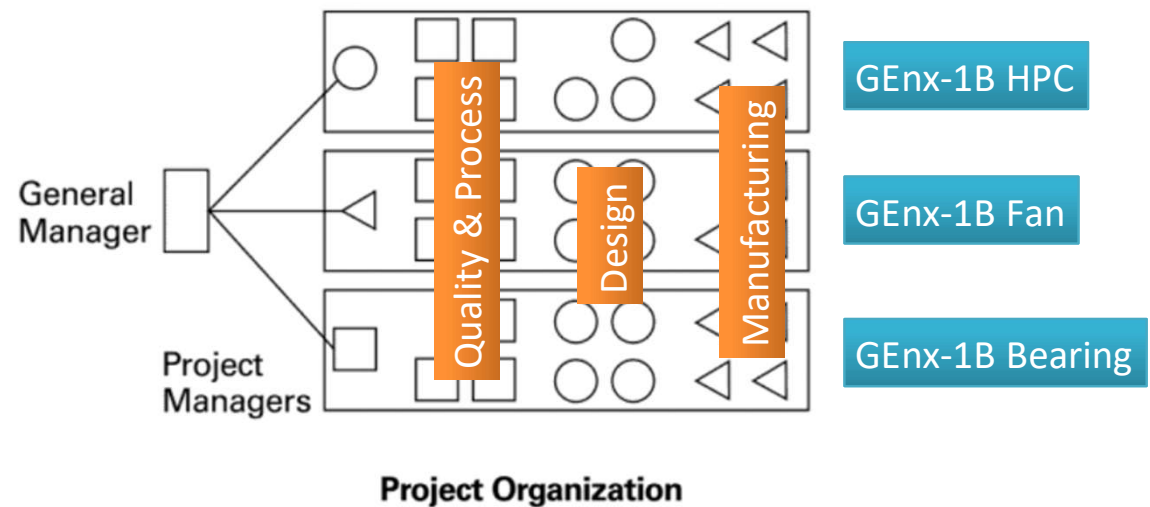


Product Development Organization

Functional and Project Organization



Grouped as per Area of Specialization



Grouped as per the Product

Functional Vs. Project Organization



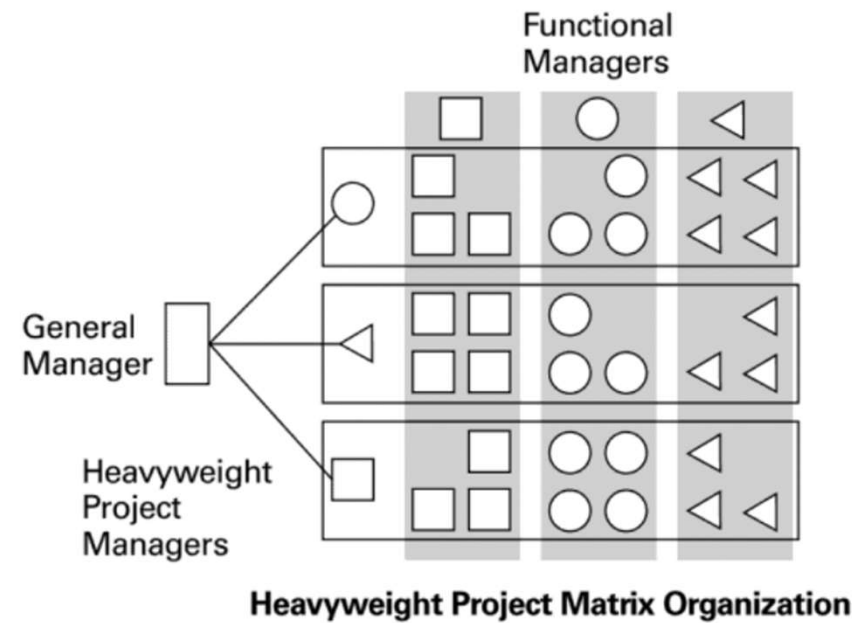
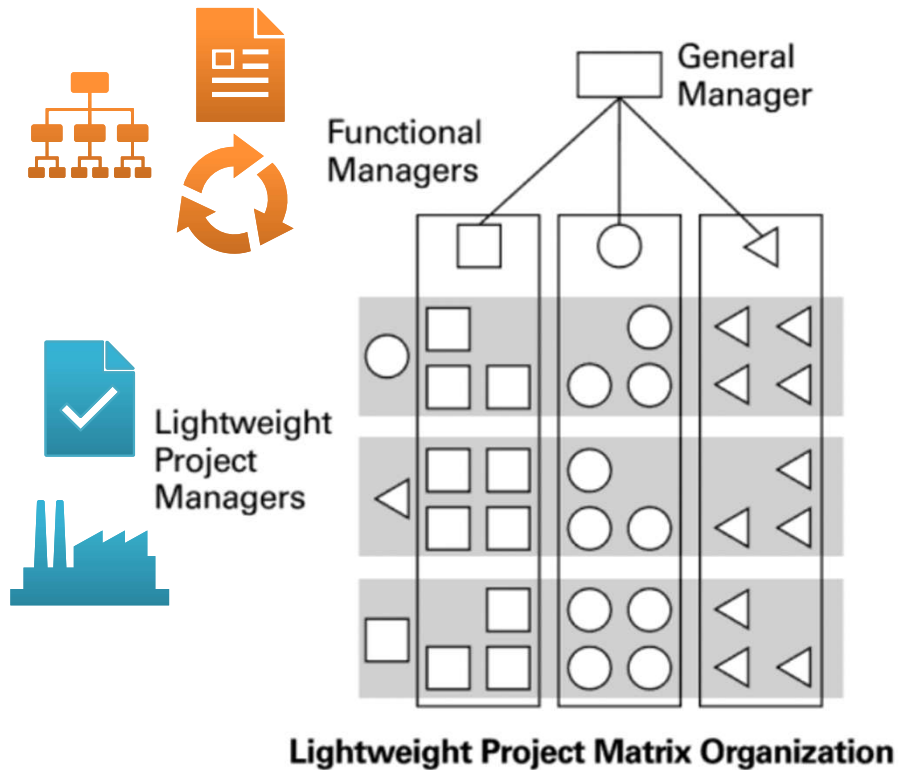
Functional Organization	Project Organization
Employees grouped by expertise, which helps the performance	Employees grouped by the 'project/product', which helps to know 'bigger picture'
Roles and responsibilities: More Accountability	More Ownership
Clear career path	Various products and product centric >>> Versatile and flexible
Well engaged department	Finding 'peers' in the team may be challenging
Feel safe and comfortable	Sense of urgency, driven by tight schedule and team setup
More job security, and hence, loyal to organization	Sense of insecurity, as the project will close once done, and hence, less loyal

Functional Vs. Project Organization



Functional Organization	Project Organization
Monotonous work >>> Lose interest	Exposure to entire PD provides learning opportunities
Effectivity of Inter-disciplinary or Inter-department or Cross-functional activities poor	The team is inter-disciplinary, and hence, more exposure
Change Management is challenging	Change Management is 'easy' to perform
Delayed decision making, and sometimes unsatisfactory/conflicting.	Faster decision making
Challenges with compiling communication and feedback	Communication and feedback are good
Resources are usually flexible, and hence, cost is managed well	Resources are sometimes 'specialized', and hence, cost can be high

Matrix Organization



Matrix Organization: Advantages



- (1) Readiness, High efficiency, and Faster to market
- (2) Usually suited for large and multi-project organizations, as the resources can be managed well
- (3) Effective management of highly skilled, experts, and capable resources
- (4) Engaged team >>> Well integrated organization
- (5) Learn and Grow <<< Opportunities to work in multiple projects

Matrix Organization: Disadvantages



- (1) Role would be confusing when you report to multiple supervisors/managers
- (2) Project Manager Vs. Functional Manager: Priority, Resource deployment >>> Affecting the employee
- (3) More 'Managers' >>> Increased Overhead cost
- (4) KPI and Career development Vs. Project Deliverables

Characteristics of Different Organizations



	Matrix Organization			
	Functional Organization	Lightweight Project Organization	Heavyweight Project Organization	Project Organization
Strengths	Fosters development of deep specialization and expertise.	Coordination and administration of projects is explicitly assigned to a single project manager. Maintains development of specialization and expertise.	Provides integration and speed benefits of the project organization. Some of the specialization of a functional organization is retained.	Resources can be optimally allocated within the project team. Technical and market trade-offs can be evaluated quickly.
Weaknesses	Coordination across different functional groups can be slow and bureaucratic.	Requires more managers and administrators than a non-matrix organization.	Requires more managers and administrators than a non-matrix organization.	Individuals may have difficulty maintaining cutting-edge functional capabilities.
Typical Examples	Customized products, where development involves slight variations to a standard design (e.g., motors, bearings, packaging).	Derivative products in many automobile, electronics, and aerospace companies.	New technology or platform projects in automobile, electronics, and aerospace companies.	Start-up companies. "Tiger teams" and "skunkworks" intended to achieve breakthroughs. Firms competing in highly dynamic markets.
Major Issues	How to integrate different functions (e.g., marketing and design) to achieve business goals.	How to balance functions and projects. How to simultaneously evaluate project and functional performance.		How to maintain functional expertise over time. How to share learning from one project to another.

How to Choose the Organization?



Factors to consider are:

- (1) Cross-functional integration
- (2) Cutting-edge functional expertise
- (3) Resource utilisation
- (4) Product development speed

Product Development Teams



The individuals can be grouped by reporting structure, business units, and physical layout.

With more modern collaboration tools, the project teams are also globally distributed, and still work effectively. The reasons include:

- (1) Access to information - regional and global
- (2) Local skills i.e. availability of technical expertise
- (3) Location of manufacturing plants
- (4) Collaboration with local suppliers
- (5) Low cost (labour and operation)
- (6) Outsourcing to increase PD capacity

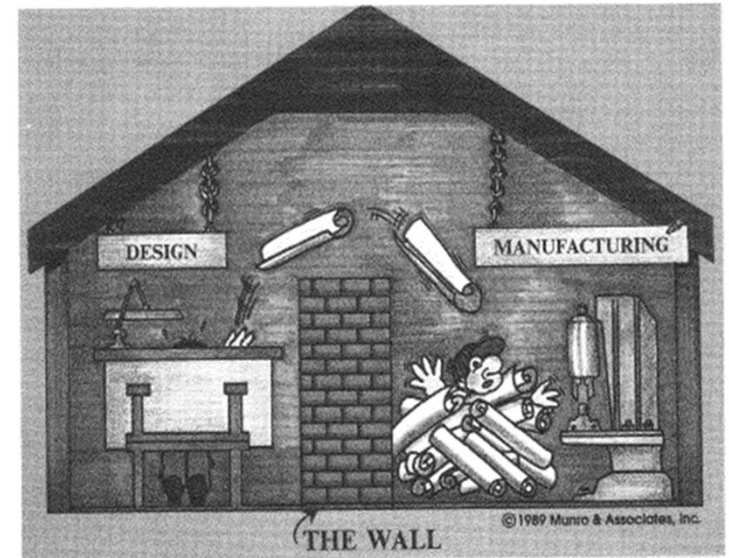


Concurrent Engineering

Conventional Way



- (1) Distinct functions
- (2) Separate departments i.e. no inter-linking



Concurrent Engineering



Key elements of concurrent engineering are:

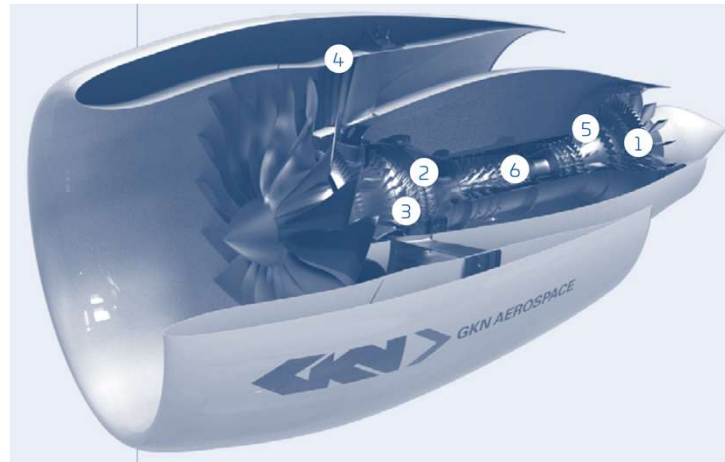
- (1) Cross-functional team >>> multi-disciplinary collaboration, increased quality
- (2) Parallel design >>> Reduce cycle time and cost
- (3) Vendor partnering
- (4) Revenue Sharing Partner (RSP), Risk and Revenue Sharing Partner (RRSP)
 - *Access to new technologies*
 - *Improved manufacturing*
 - *Lower development costs*
 - *Knowledge exchange*

RRSP: Melrose (GKN Aerospace)



Melrose's RRSP contribution focuses primarily on the design, manufacture and support of engine mount structures and cases, which typically last the life of the engine.

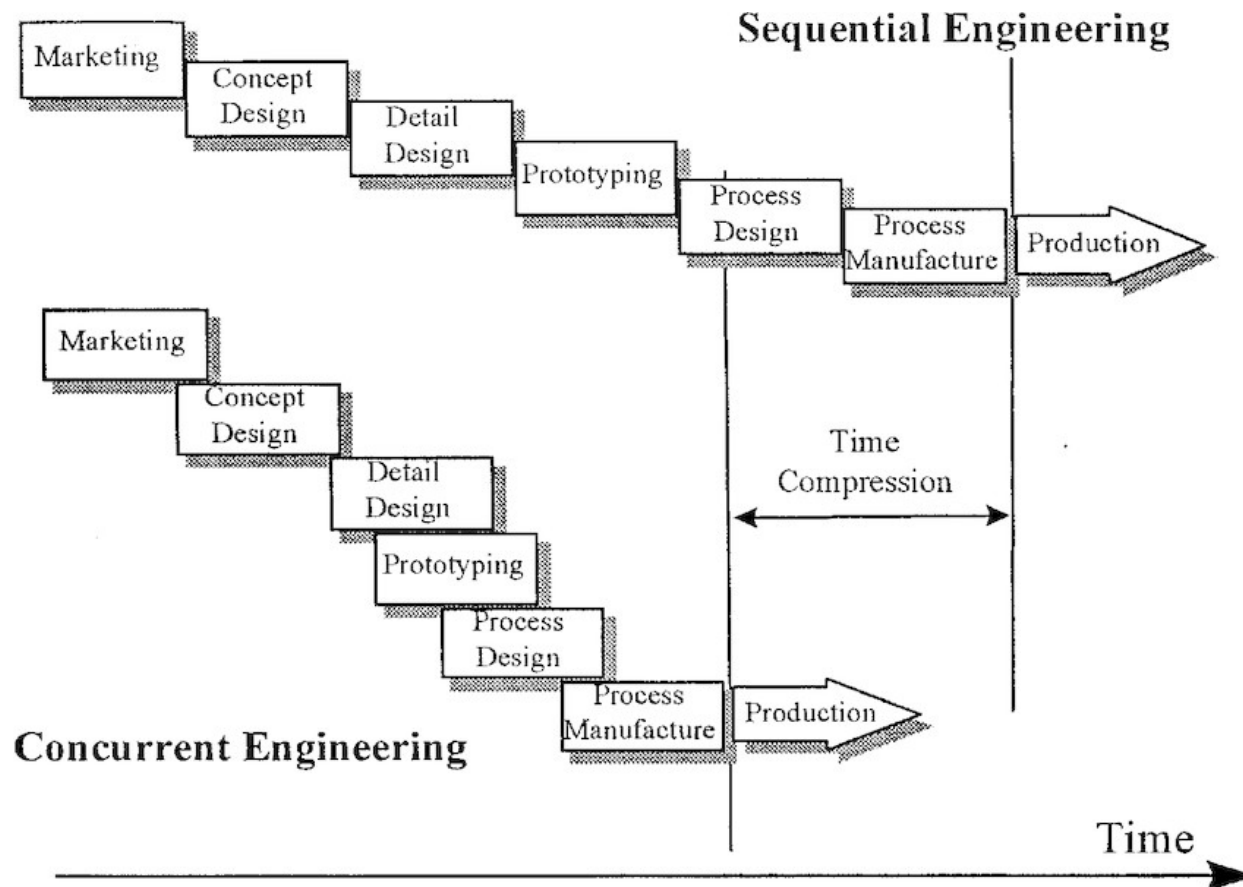
- (1) Contracts lasting for 50 years (life-of-programme)
- (2) 19 RRSP contracts



We are bringing expertise in the design of complex load-carrying structures, as well as in the use of our proprietary additive fabrication with Laser Wire Deposition (LWD) technology.

Ref: <https://www.melroseplc.net/media/p23g1rby/melrose-rrsp-booklet-digital.pdf>

Concurrent Vs. Sequential Engineering





Introduction to Product Planning

Introduction to Product Planning



- The product planning process takes place before a product development project is formally approved.
- Product planning is an activity that considers both the current product line and the potential portfolio of projects that an organization might pursue.
- It also determines what subset of the 'possible' development projects will be pursued over what time period.

Introduction to Product Planning



Questions to be addressed:

- (1) What product development projects will be undertaken?
- (2) What mix of fundamentally new products, platforms, and derivative products should be pursued?
- (3) How do the various projects relate to each other as a portfolio?
- (4) What will be the timing and sequence of the projects?

Introduction to Product Planning



The PD team needs to know its mission before beginning development.

- (1) What **market segments** should be considered in designing the product and developing its features?
- (2) What new **technologies** (if any) should be incorporated into the new product?
- (3) What are the manufacturing and service **goals and constraints**?
- (4) What are the **financial** targets for the project?
- (5) What are the **budget and time frame** for the project?

Product Planning Process



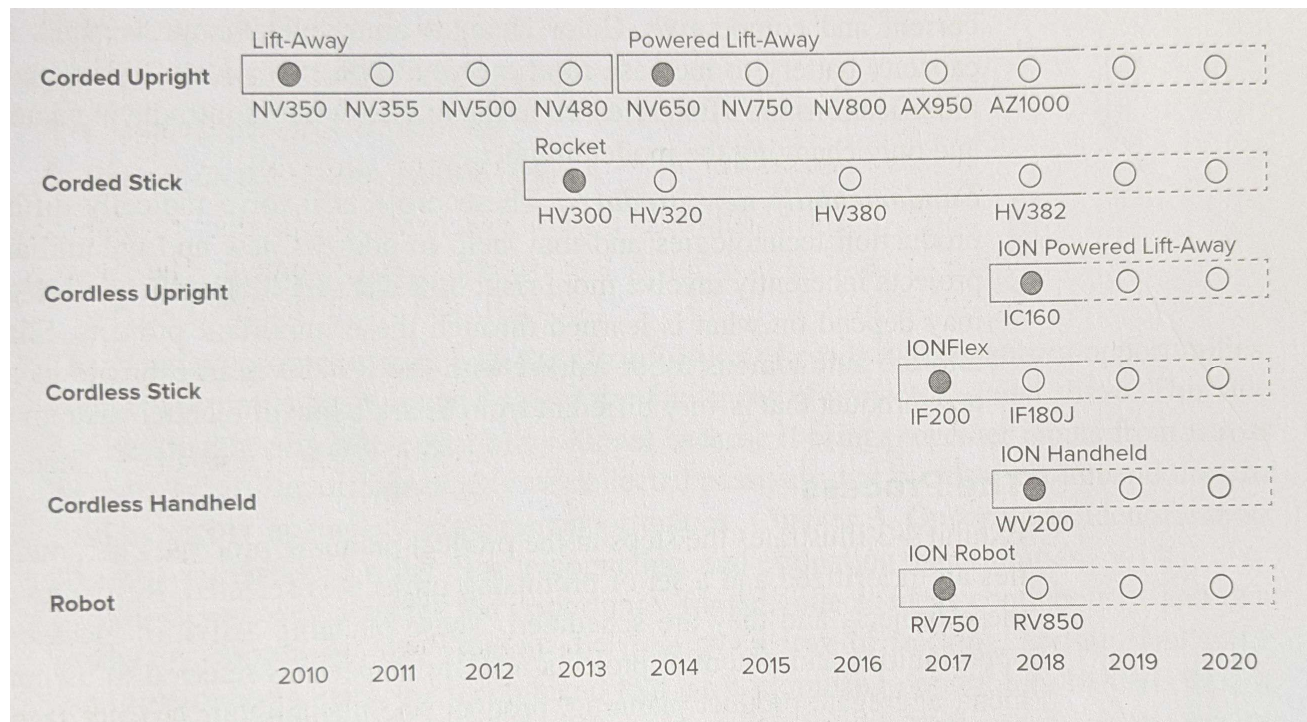
Key aspects of Product Planning (PP) process

- Portfolio of the products to be developed and the timing
- Considers PD opportunities
- Adopt changes in the competitive environment, technology, and field data

Product Planning Process: Example



Example: Shark brand vacuums

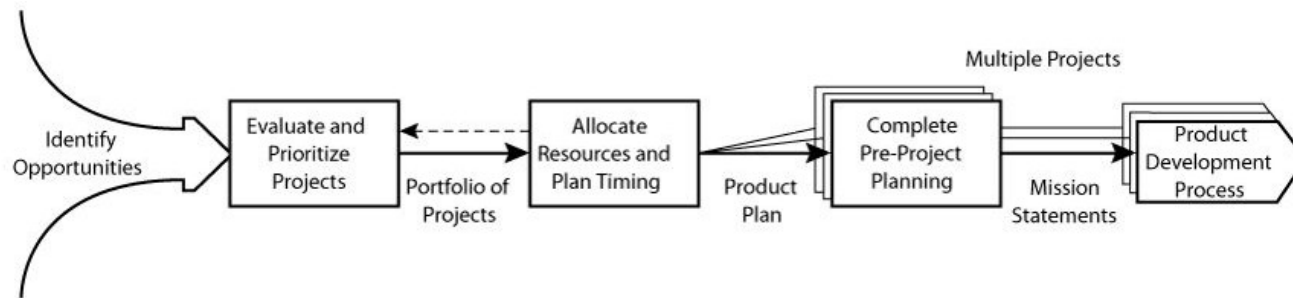


Types of PD Projects



- (1) **New product platforms:** New product based on a new, common platform. It requires major development effort.
- (2) **Derivatives of existing product platforms:** These projects extend an existing product platform to address familiar markets with new product or feature.
- (3) **Incremental improvements to existing products:** Modifying some of the features of existing products in order to keep the product line current and competitive (cost, servicability, etc.).
- (4) **Fundamentally new products:** These projects involve radically different product or production technologies and may help to address new and unfamiliar markets.

Product Planning Process



From *Product Design and Development* by Karl Ulrich and Steven Eppinger (McGraw-Hill/Irwin)

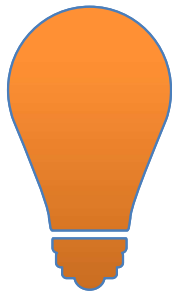
1. Identify opportunities.
2. Evaluate and prioritize projects.
3. Allocate resources and plan timing.
4. Complete pre-project planning.
5. Reflect on the results and the processes.

Ability to adjust the product plan is VITAL to the success!



#1 Identify Opportunities

#1: What is an Opportunity?



An idea for a new product!

- New need
- New technology
- New solution

#1: Type of Opportunities

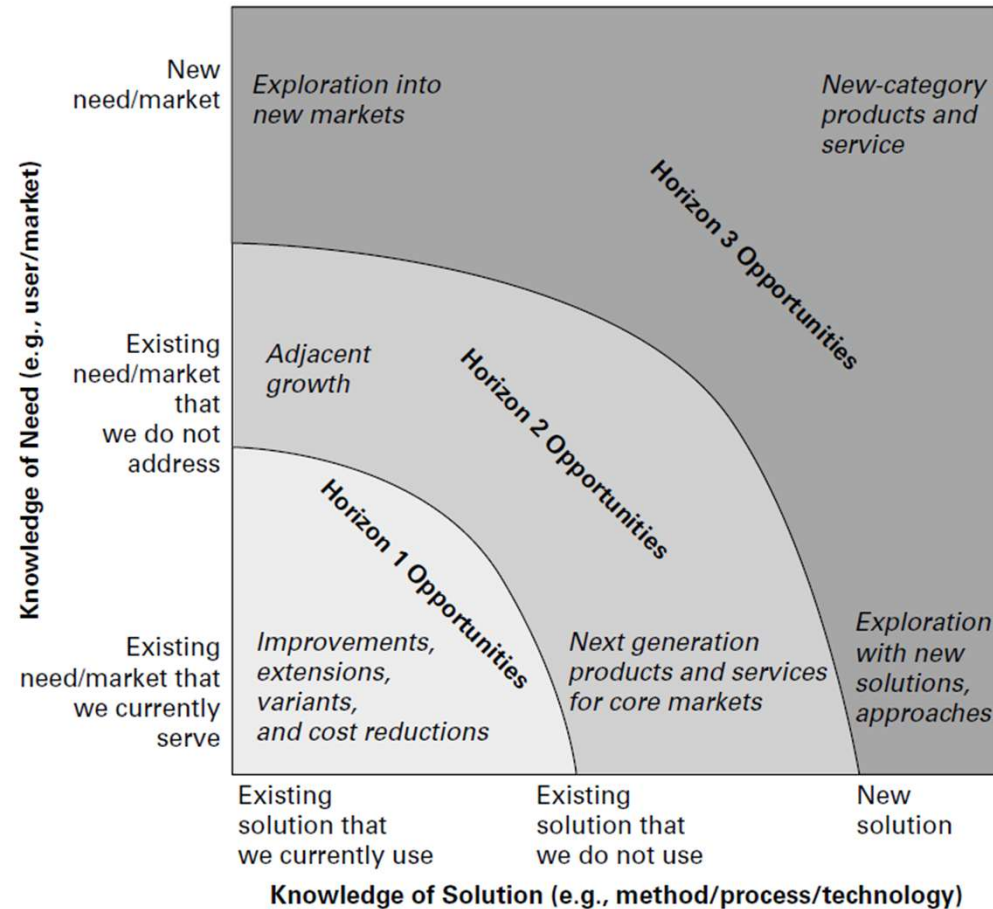


Opportunities can be categorized based on:

- What the team knows already (solution or need)
- Uncertainty

Horizon 1	Low Risk	Existing Products	Improvement, variants, cost reduction
Horizon 2	Medium Risk	Combination	Combination
Horizon 3	Hight Risk	New to world	New technology or product

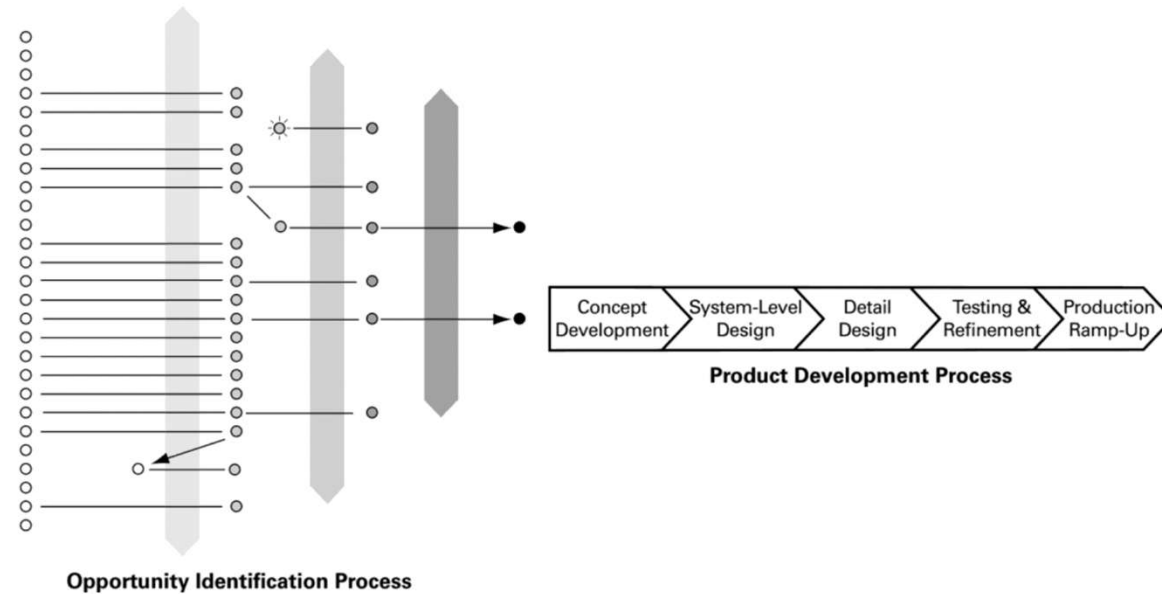
#1: Type of Opportunities



#1: Opportunity Identification Process



- (1) Establish a charter
- (2) Generate and sense many opportunities
- (3) Screen opportunities
- (4) Develop promising opportunities
- (5) Select exceptional opportunities
- (6) Reflect on the results & the process



Large number of Opportunities

High variance

High quality

#1: Opportunity Identification Process



“Broader” charter

Study and listen to the Customers

Passion helps

Trends

Annoyance gives birth to opportunity

External source of ideas

Exploit the resources

Most votes Vs. Strong vote

Uncertainties and cost

Real-Win-Worth-it



#2 Evaluate and Prioritize Projects

#2 Evaluate and Prioritize Projects



Four basic perspectives for new products in existing product categories:

- (1) Competitive strategy
- (2) Market segmentation
- (3) Technological trajectories
- (4) Product platforms

#2.1: Competitive Strategy



(1) Technology Leadership

>>> *Great emphasis on R&D of new technologies, and its deployment through PD*

(2) Cost Leadership

>>> *Focus on efficiency, superior manufacturing method, low-cost operation*

(3) Customer Focus

>>> *Focus on customer need and preferences*

(4) Imitative

>>> *Follow the market and competitors*

#2.2: Market Segmentation



Map competitor's products and the firm's own products onto segments.

Assess the opportunities by identifying the weaknesses in its own product line and in the offerings of the competitors.

#2.3: Technology Trajectories



Answer to the question “When to adopt a new basic technology in a product?”

The S-curve highlights how AM technology matures from initial R&D (lower, flatter part of the curve) to high-volume production, reducing weight and improving engine performance



Pioneering Additive Manufacturing

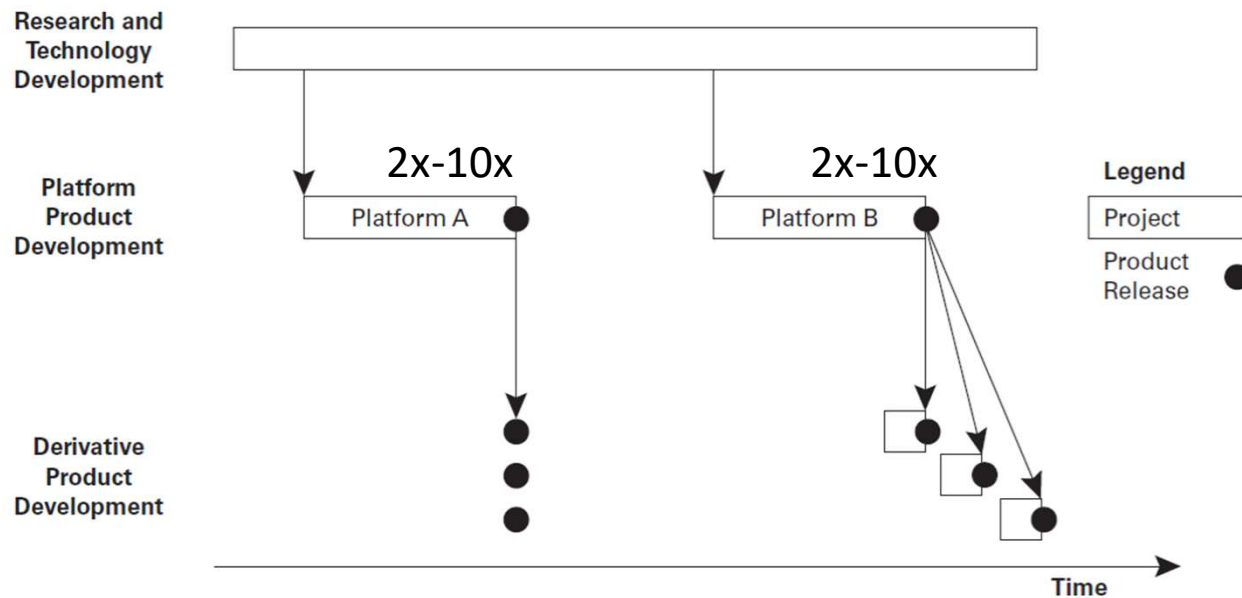
We've been pioneering the use of Additive Manufacturing for two decades and have our AM components flying on multiple platforms. Our innovative material solutions involve a process called fabrication, which is like slicing a cake into smaller pieces and then putting them back together in a new shape. This method allows us to transform large and complex geometries into high-performance parts. These efforts are all part of our commitment to making the aviation industry more efficient, which benefits both our business and the environment.

<https://www.gknaerospace.com/innovation/additive-manufacturing/>

#2.4: Product Platform Planning



Usage of existing platforms to introduce a product rapidly, easily, and economically.



#2: Technology Roadmapping



A way to represent the expected availability and future use of various technologies, relevant to the products being considered.

GE Aviation's Roadmap for the Future of Flight

July 14, 2022 | by Chelsey Levingston

Share



Over the last year, GE Aviation has outlined what is one of the company's most extensive technology development roadmaps in its more than 100-year history. Multiple ground and flight tests planned this decade will seek new, breakthrough technologies for use in next-generation commercial aircraft engines that could enter service beginning in the mid-2030s.

Advanced engine architectures such as open fan, hybrid electric propulsion systems, and new compact engine core designs—just some of the technologies that will be demonstrated on test stands over the coming years—will all be key programs to watch in 2022 and beyond.

In addition to maturing these technologies for flight readiness and new products, GE also supports efforts to increase use and availability of alternative fuels, such as Sustainable Aviation Fuel (SAF) and hydrogen.

Revolutionary technologies and alternative fuels both have critical roles to play in [meeting the aviation industry's long-term climate goal](#) of net-zero carbon emissions by 2050 for commercial flight.

Here, we've laid out GE's top innovations and industry-leading efforts to watch in the march toward net-zero.

03/15/2021 · Press Release

Power Day: Volkswagen presents technology roadmap for batteries and charging up to 2030

Ref: <https://www.geaerospace.com/news/articles/farnborough-airshow-sustainability-technology/ge-aviations-roadmap-future-flight>



#2: Evaluating Fundamentally New Product Opportunities

It is necessary for the firm to invest in new technologies or new market for sustainable growth, though it is risky and involve cost/resource. Some of the criteria for evaluating such opportunities are:

- Market demand and growth rate
- Sustainability
- Knowledge and Awareness about the market and technology
- Current product portfolio and the capabilities
- Potential for patents, trade secrets, or other barriers to competition

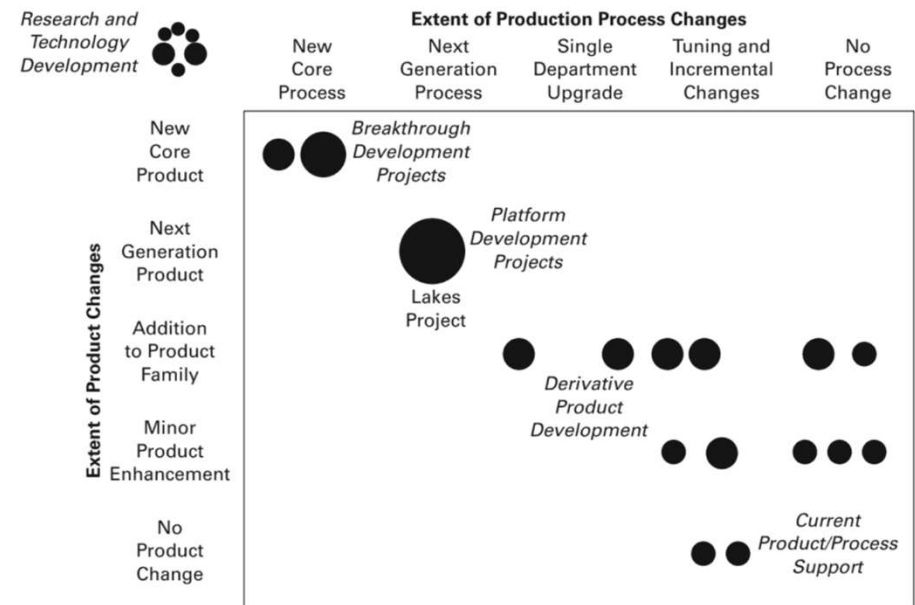
#2: Balancing the Portfolio



Organization's portfolio are managed by mapping the portfolio using dimensions such as technical risk, financial return, market need, etc.

Other mapping includes 'extent of change in product line' and 'extent of change in production process'

Organization often 'trade-off' investments of different types



Adapted from Wheelright and Clark, 1992



#3 Resource Allocation and Plan Timing



#3: Allocate Resources and Plan Timing

Allocated budget Vs. Promising projects Vs. Limited resources

#3.1: Resource Allocation



Too many projects without regard for the availability of resources would result in skilled engineers and managers allotted to too many projects.

This would cause:

- Drop in productivity >> Delayed deliveries >> Late to market >> Loss in profit!
- Stressful job >> Dis-engaged >> Loss of resources!

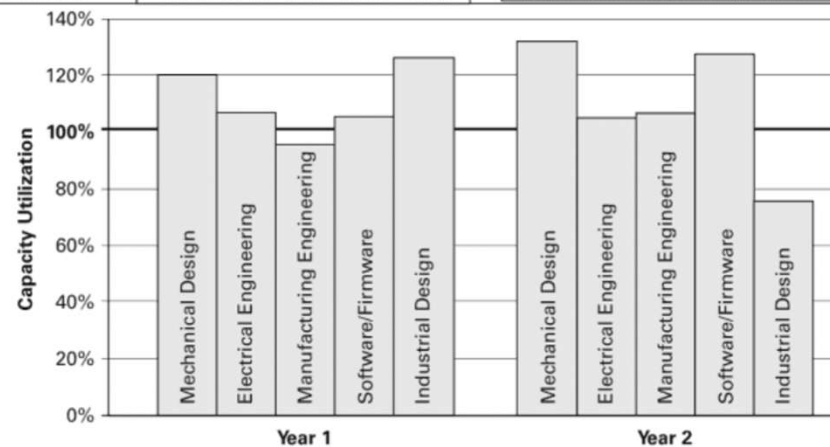
Ideal is not to deploy a resource in more than three projects.

#3.1: Resource Allocation



Example showing demand Vs. capacity

	Year 1					Year 2				
	Mechanical Design	Electrical Engineering	Manufacturing Engineering	Software/Firmware	Industrial Design	Mechanical Design	Electrical Engineering	Manufacturing Engineering	Software/Firmware	Industrial Design
Lakes Project	155	160	105	75	7	210	160	140	80	4
6010 Project	30	25	10	5	1	25	20	5	6	
595 Project	60	24	25			20	15	15		
Astro Project	55	60	44	25	2	75	65	50	40	2
Resource Demand	300	269	184	105	10	330	260	210	126	6
Resource Capacity	250	250	200	100	8	250	250	200	100	8
Capacity Utilization	120%	108%	92%	105%	125%	132%	104%	105%	126%	75%



#3.2: Project Timing



Timing and sequence of projects must consider:

- Timing of product introduction: Quality Vs. Speed
- Technology readiness: Robustness
- Market readiness: When to launch? When to improve?
- Competition: Know the competition



#4 Complete Pre-Project Planning



#4: Complete Pre-Project Planning

Core team, a small and cross-functional team, formulates

- (1) Product Vision Statement (very general)
- (2) Mission Statement (detailed and more specific)



#4: Complete Pre-Project Planning

Mission Statement may include:

- Basic function of the product
- How it would benefit the customer
- Business goal, including time, cost, quality
- Primary and secondary markets
- Assumptions and constraints. E.g. Manufacturing, Service, and Sustainability
- List of stakeholders



#5 Reflect on the Results and the Process



#5: Reflect on the Results and Process

A reality check is required before proceeding with the PD process, else the consequence would be severe and expensive!

Questions can be associated with:

- Opportunity funnel
- Product plan supporting firm's strategy and opportunities
- Resource allocation
- Creativity
- Acceptance of mission statement
- Over constrained objective
- Improvement opportunities



Summary

Summary



- (1) Discussed various types of PD organization. What kind of organization to choose depends on various factors like (a) Need of cross-functional integration, (b) Securing cutting-edge functional expertise, (c) Resource utilisation strategy, and (d) Speed of PD
- (2) Discussed the importance of concurrent engineering in the current industrial trends
- (3) Discussed the key aspects of Product Planning, different types of PD projects, and steps involved in PP process
- (4) Highlighted the role of “Finding Opportunities”, and how crucial to arrive at a good “Mission Statement”



Thank You!



BITS Pilani

Pilani | Dubai | Goa | Hyderabad

Product Design

(AE ZG531 / AEL ZG531 / DE ZG541 / DM ZG541 / MM ZG541)



BITS Pilani
Pilani | Dubai | Goa | Hyderabad

Lecture-3: Identifying Customer Requirements

Mission Statement Template



Product Description:

- One concise and focused one sentence

Key Business or Humanitarian goals:

- schedule
- gross profit /margin.
- market share.
- Advancement of human needs

Primary Market:

- Brief phrase of market sector or group

Secondary Market:

- List of secondary market currently or perceived

Assumption :

- Key assumptions or uncontrolled factors to be confirmed by customer

Stake Holders:

- 1-5 word statement of customer set

Avenues of compact design:

- Identify key areas of innovation

Scope Limitation:

- List of limitation that will reign back the design team from 'solving the world'

Technical Questioning: Example



- What is the problem really about?
- What implicit expectations and desires are involved?
- Are the stated customer needs, functional requirements, and constraints truly appropriate?
- What avenues are open for creative design and inventive problem solving?
- What avenues are limited or not open for creative design? Limitations on scope?
- What characteristics/properties must the product have?
- What characteristics/properties must the product not have?
- What aspects of the design task can and should be quantified now?
- Do any biases exist with the chosen task statement or terminology? Has the design not been posed at the appropriate level of abstraction?
- What are the technical and technological conflicts inherent in the design task?

Technical questioning for fingernail clipper



- What is the problem really about?
 - Clumsy operation of a typical clipper.
- What implicit expectations and desire are involved?
 - Remain a manual clipper that can be operated by oneself.
- Are the stated customer needs, functional requirements and constraints truly appropriate?
 - Many groups, as documented in trade magazines and customer studies, have noted clumsy operation. A detailed assessment will be made in post-customer interviews.

Technical questioning for fingernail clipper



- What avenues are open for creative design?
 - Can modify any and all parts. Introduce different materials, combine the functional parts and many more....
- What are the limitations for creative design?
 - No electrical power, no service based solution.
- What properties / characteristics must the product have ?
 - Easy to use, durable, safe, low cost

Technical questioning for fingernail clipper



- What properties / characteristics must the product not have ?
 - Should not be bulky.
- What aspect of the design task can and should be quantified now?
- Do any biases exist with the chosen task statement or terminology?
- Has the design task been posed at the appropriate level of abstraction?
- What are the technical and technological conflicts inherent in the design task?

Mission statement: Fingernail clipper



Mission Statement: Fingernail Clipper Product	
Product Description	Remove and file excess fingernail length.
Key Business or Humanitarian Goals	• 6 month development of beta prototype • 30% profit margin • Initial 5% market share • Supplement fingernail polish business
Primary Market	• Adults of all ages, focusing on fingernail polish users
Secondary Market	• Knife collectors • Business executive
Assumptions	• Small, compact stowage volume • Long life (10-20 years)
Stakeholders	• XXX Corporation; user; salons, retailers
Avenues for Creative Design	• ergonomic shape; store/capture of nails; compact stowage; ease of cutting
Scope Limitations	• Materials: steel processing and moldable plastics.

Mission statement: Example



Mission Statement: Multifunctional Office Document Machine	
Product Description	• Networkable, digital machine with copy, print, fax, and scan functions
Benefit Proposition	• Multiple document processing functions in one machine • Connected to office computer network
Key Business Goals	• Support Xerox strategy of leadership in digital office equipment • Serve as platform for all future B&W digital products and solutions • Capture 50% of digital product sales in primary market • Environmentally friendly • First product introduction 4th Q 1997
Primary Market	• Office departments, mid-volume (40–65 ppm, above 42,000 avg. copies/mo.)
Secondary Markets	• Quick-print market • Small “satellite” operations
Assumptions and Constraints	• New product platform • Digital imaging technology • Compatible with CentreWare software • Input devices manufactured in Canada • Output devices manufactured in Brazil • Image processing engine manufactured in both the United States and Europe
Stakeholders	• Purchasers and users • Manufacturing operations • Service operations • Distributors and resellers

Mission Statement

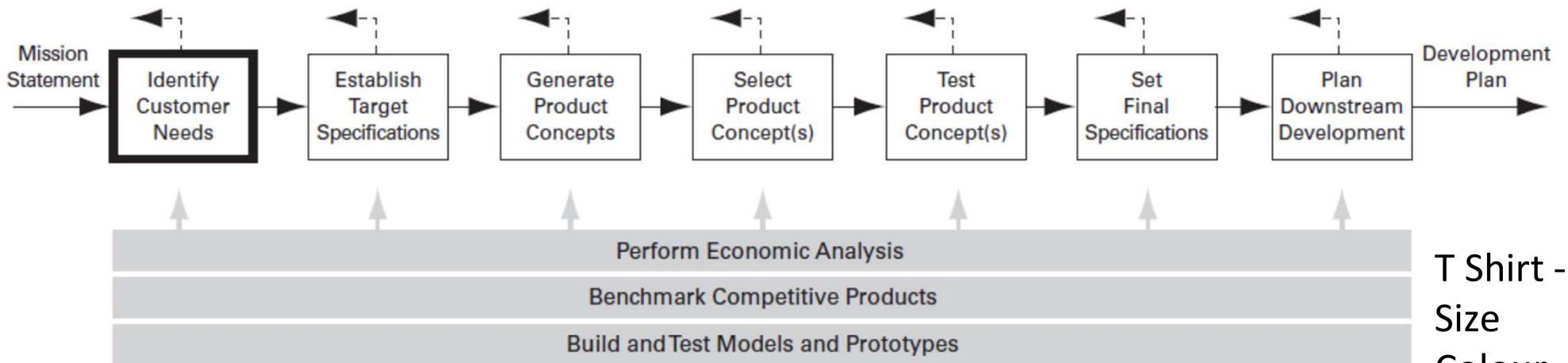


Mission Statement: Thermostat Project

Product Description	• Thermostat for residential use.
Benefit Proposition	• Simple to use, attractive, and saves energy.
Key Business Goals	• Product introduced in fourth quarter of 2012. • 50% gross margin. • 10% share of replacement thermostat market by 2016.
Primary Market	• Residential consumer.
Secondary Markets	• Heating, ventilation, and air conditioning contractors doing residential work.
Assumptions	• Replacement for an existing thermostat. • Compatible with most existing systems and wiring.
Stakeholders	• User • Retailer • Sales force • Service center • Production • Legal department

Mission Statement specifies the direction but not a precise destination or a particular way to proceed

Customer Need Identification Steps



T Shirt -
Size
Colour
Material
Sleeves
cost

1. Gather raw data from customers.
2. Interpret the raw data in terms of customer needs.
3. Organize the needs into a hierarchy of primary, secondary, and (if necessary) tertiary needs.
4. Establish the relative importance of the needs.
5. Reflect on the results and the process.

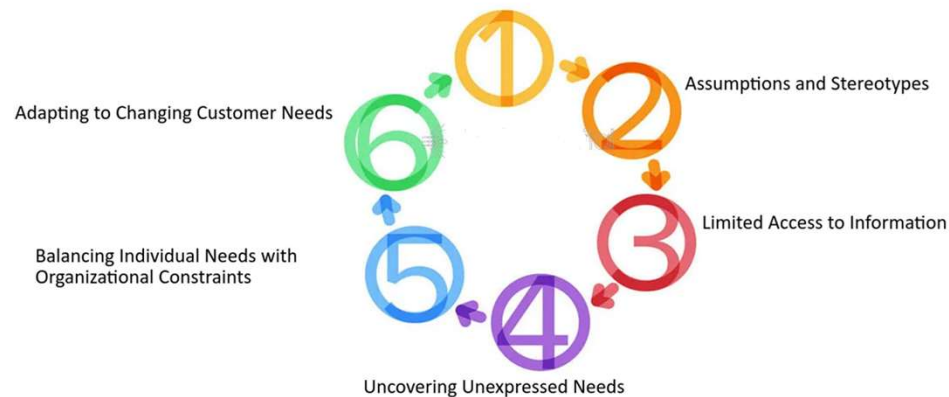
Challenges in Customer Need Identification



Customer **NEEDS** keep CHANGING



Lack of Communication



Example: Customer's Criteria for Shoe



Top Shoe Criteria?

1. Fit/comfort
2. Price
3. Brand
4. Functional purpose
5. Color availability
6. Selection
7. Product availability
8. Promotions
9. Sustainability
10. Silhouettes/styles
11. In-store experience
12. Replacement
13. Online experience
14. Personalization
15. Customization
16. Ethically sourced



<https://www.simon-kucher.com/en/insights/footwear-purchasing-criteria>

Customer Focus in Company's Values



OUR 6 VALUES

Chosen by our
teams, embodied
by our teams

Because the way we work matters as much as what we produce, we have grown our company on the ground of 6 core values that guide the behaviours of Airbus employees around the world.

Customer focus, integrity, respect, creativity, reliability and teamwork.

These values shape the DNA of our business, reflecting the unique blend of qualities found in every member of our global family.

AIRBUS

Customer Focus

Customer focus is deeply embedded into our mindset, driving us to deliver on time, on cost and on quality. We partner with our customers to satisfy their needs and deliver benefits through valuable and sustainable solutions.

We understand that each of us has an impact on customer relations through the work we do, which is why we always act with integrity and to the highest professional standards.

<https://www.airbus.com/en/careers/our-values>

Customer Engagement



Tata Motors launches its annual customer-engagement program, Grahak Samvaad, for its customers across India

The nationwide commercial vehicle **customer-engagement program** will focus on enhancing the overall customer experience

Key highlights:

- Grahak Samvaad to be held from 20th to 28th October 2021, to educate the customers about the company's innovative service and product offerings
- October 23rd is designated as **National Customer Care day** to commemorate the delivery of the first Tata Motors truck in 1954

Mumbai, 20th October 2021: Tata Motors, India's largest commercial vehicle manufacturer, will celebrate 'National Customer Care Day' on 23rd October to commemorate the day when the first truck was rolled out from Tata Motors Jamshedpur plant in 1954. The company will also launch its annual customer-engagement program, the Grahak Samvaad, from 20th to 28th October 2021. The program aims to educate the customers about the company's innovative service and product offerings. Executives from Tata Motors will interact with customers **to collect feedback and understand their expectations, key pain points and suggestions to further streamline its aftersales service and improve its product offerings for a hassle-free driving experience.**

Commenting on this unique program, **Mr. R Ramakrishnan, Global Head – Customer Care, Commercial Vehicle Business Unit, Tata Motors**, said, "Grahak Samvaad acts as one of the major differentiators for customers opting for commercial vehicles from Tata Motors. The exemplary aftersales service ensures maximum uptime and lower total cost of operations (TCO) throughout the vehicle's entire life cycle. National Customer Care Day, every year presents us with a **golden opportunity to converse with customers, gather precious feedback for our initiatives, that helps us further improve the quality of our aftersales service and customer relationship.** We look forward to the suggestions, views and market information received via our partners and customers and implement those learnings in the development of our products and services."

<https://www.tatamotors.com/wp-content/uploads/2023/11/press-20oct21.pdf>

Industry Shift Vs. Customer Need



TATA MOTORS

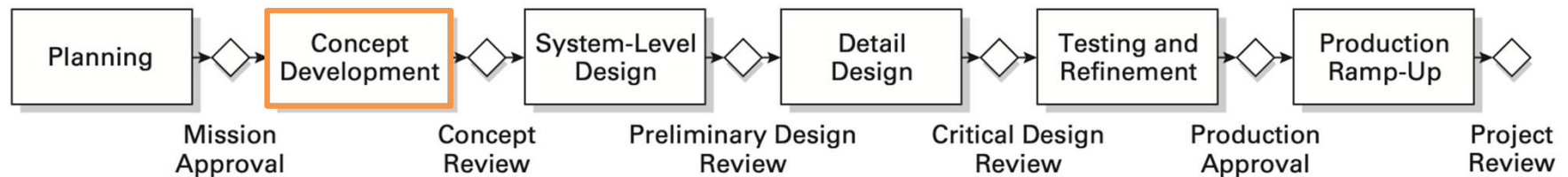
<https://www.tatamotors.com/blog/constructing-contemporary-vehicles-global-design-ethics-and-increasingly-design-literate-indian-consumers/>

The industry-wide shift towards electric mobility and clean, emission-free vehicles is also definitely impacting design. One must reimagine what vehicles with alternative powertrains would look like. This must be in line with the needs of the customers. As mentioned before, the best design is one that creates a stimulation within the customer as well as makes them comfortable . Even beyond electrification,

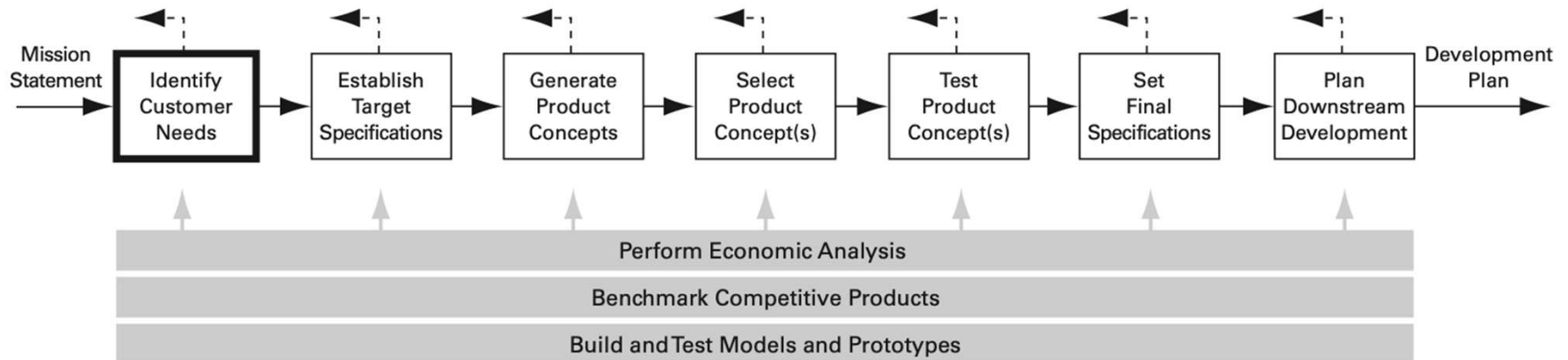
When Does it Happen?



Generic PD Process



Concept Development



What is a Latent Need?



‘Needs’ can be of two types

- (1) Expressed Needs - What customers say they want
- (2) Latent Need - What customer is unaware of

E.g. Camera in mobile phones by Nokia and Motorola

Paying special attention to what customers do, not what they say, is key to the successful development and marketing of new products

If fulfilled, then it would result in greater customer satisfaction.

Recognising Latent Needs is critical in Product Development

<https://www.washington.edu/news/2004/10/11/successful-development-of-new-products-requires-anticipating-customers-needs/>



#1: Gather 'Raw' Data from Customers

Step 1 - Gathering Information from Customers



Customer Complaint Form	
_____ record all customer complaints that they _____	

_____ item (use a CAP where applicable) of resolution	
_____ y be used though other records are acceptable as well.	
Complaint	
Date:	_____
Corrective Action	

Customer Communication	
Date:	_____

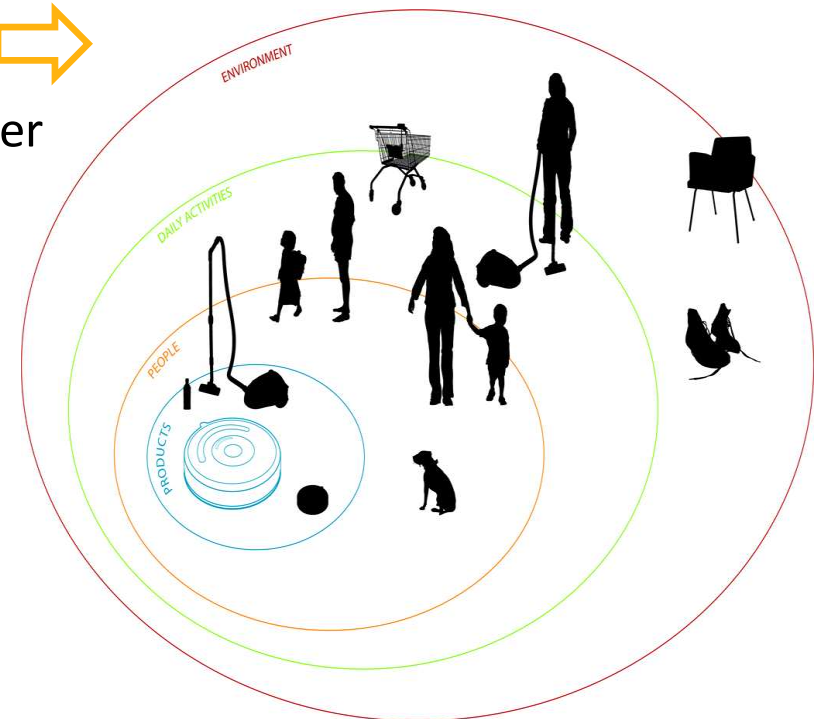


Customer Survey	
Please rate the following areas so that we may improve our service	
Quality of service	Good Average Poor Ve
Excellent	☐ ☐ ☐ ☐
Quality of items sup.	
Excellent	☐ ☐ ☐ ☐

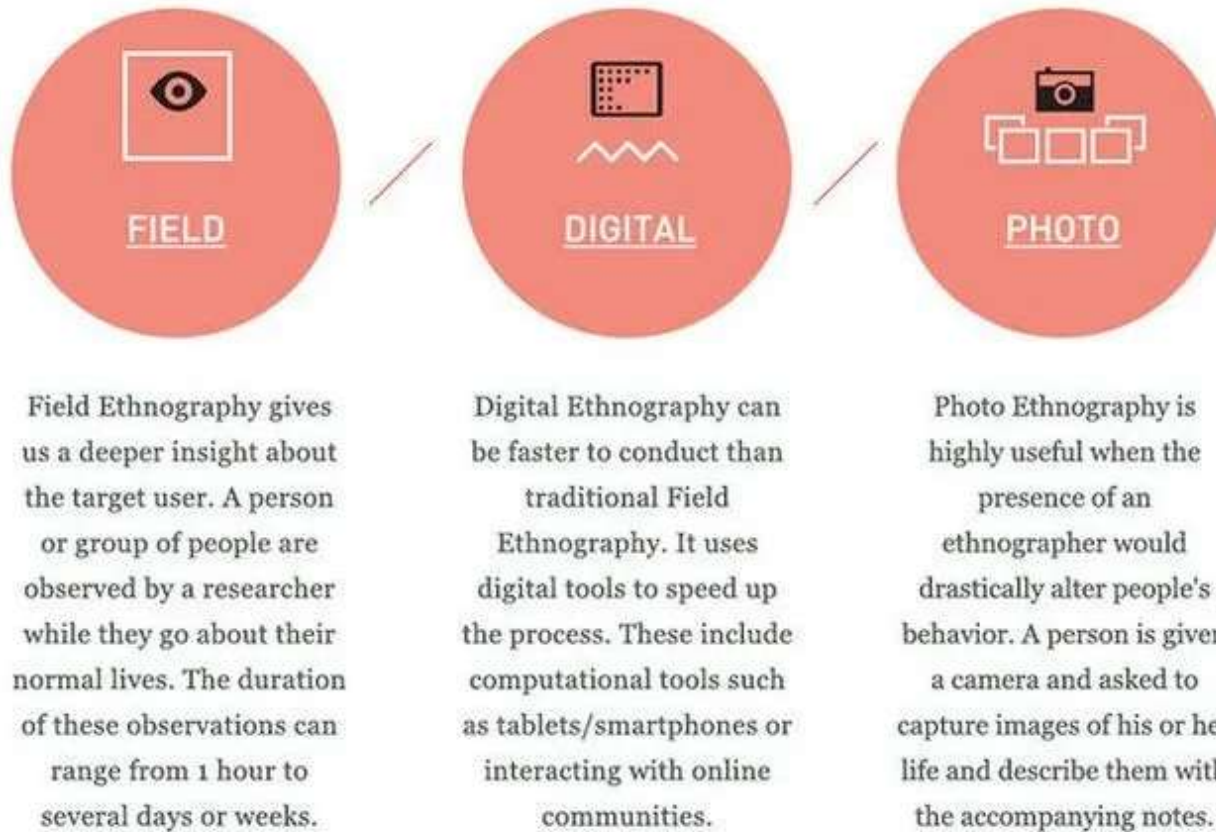
Ethnographic Studies



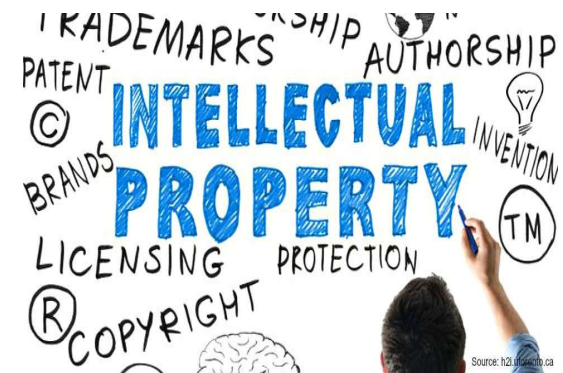
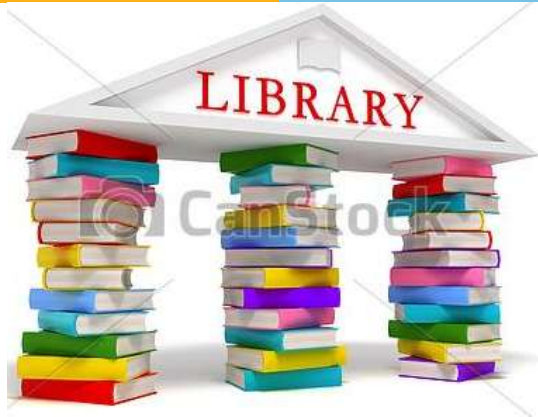
- Way people behave in their regular environments
- How a customer uses (or misuses) a product
- Actual end users interacting with the product under typical use conditions
- Way of conducting the study:
 - Video
 - Photographs
 - Sketches
 - Interview



Types of Ethnographic Studies



Other Important Sources of Information Gathering





#1: Methods to Gather Data from Customers

It is important to create a high-quality information channel directly from the customer.

Commonly used methods to gather data from customers are:

(1) **Interviewing** customers individually

(2) **Group discussion** with customers

(3) **Observing** customers using the product or performing the intended task e.g. UX sessions, Beta events



Efficient and less costly!

Other methods: Surveys or Surveys combined with above methods



Receptive

Elicit expression of needs

Open mind

Product/Visual demo



Confront

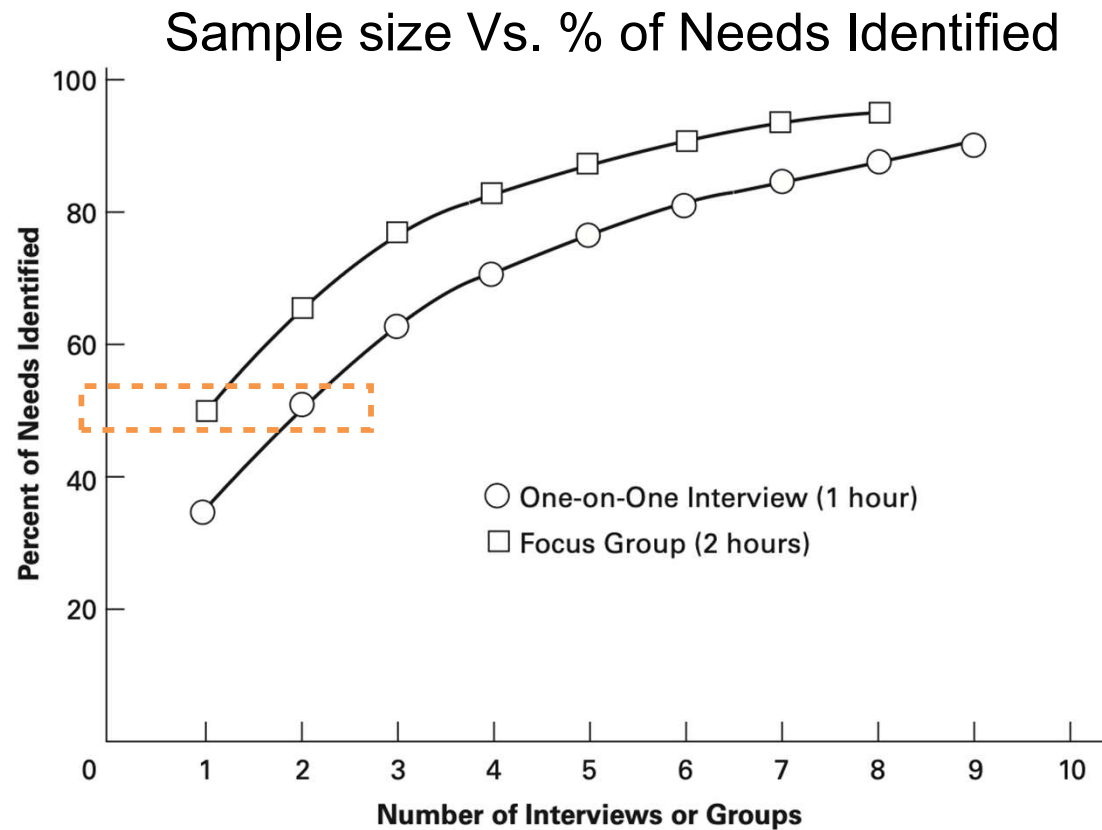
Adhere to the 'guide'

Defensive posture

Sales motive

Convince

#1: Choosing Sample Size



#1: Choosing 'Type' of Customer



Buyer



Lead User



Extreme User



End User

Experienced the pain ahead
Articulate emerging needs
Already found the solution

Unusual way
Special needs
Reveal latent needs

<https://www.flaticon.com/>

Constructing a Survey Instrument



Determine the survey purpose

Determine the type of data-collection method to be used

Identify what specific information is needed

Design the questions appropriately (unbiased)

Do not use jargon or sophisticated vocabulary

Use simple sentences

Do not lead the customer toward the answer you want

In any list of options given to the respondents, include the choice of “Other” with a space for a write-in answer

Always include one open-ended question

The number of questions should be such that they can be answered in about 15 - 30 minutes

Be sure to include instructions for completing and returning it

#1: Interviewing Techniques (Questions)



Questions to ask and Questions to provoke...

When and Why use the product?

Like and dislike

Walk us through the usage

Improvement ideas

Issues

Document the outcome as soon as possible!



#2: Interpret 'Raw' Data in Terms of Customer Needs

#2: Raw Data → Customer Needs



Example:

Question/Prompt	Customer Statement	Interpreted Need
Typical uses	I have to manually turn it on and off when it gets too hot or cold.	The thermostat maintains a comfortable temperature without requiring user action.
	Each time I want to change the temperature, I need to adjust both thermostats in the house.	Any user inputs need not be made in multiple locations. (!)
Likes—current model	I like that I can change the temperature if the setting is too high.	The temperature setting is easy to control manually.
	It didn't cost a fortune.	The thermostat is affordable to purchase.
Dislikes—current model	I'm too lazy to learn how to figure it out.	The thermostat requires little or no user instruction or learning.
	I sometimes forget to turn it off when we leave the house.	The thermostat saves energy when no one is home. (!)
	Sometimes the buttons don't register a push and I have to press it repeatedly.	The thermostat definitively registers any user inputs.
Suggested improvements	I would like my iPhone to adjust my thermostat.	The thermostat can be controlled remotely without requiring a special device.
	I like to shift quickly between different options like energy saving or ultra comfort.	The thermostat responds immediately to differences in occupant preferences.

#2: Guidelines to Translate



Guideline	Customer Statement	Needs Statement— Right	Needs Statement— Wrong
"What" not "How"	I would like my iPhone to adjust my thermostat.	The thermostat can be controlled remotely without requiring a special device.	The thermostat is accompanied by a downloadable iPhone app.
Specificity	I want a cooling system.	The thermostat can control heating and cooling systems.	The thermostat is versatile.
Positive not negative	I get frustrated standing in front of my thermostat to program it.	The thermostat can be programmed from a comfortable position.	The thermostat does not require me to stand in front of it for programming.
An attribute of the product	I have to manually override the program if I'm home when I shouldn't be.	The thermostat automatically responds to an occupant's presence.	An occupant's presence triggers the thermostat to automatically change modes.
Avoid "must" and "should"	I'm worried about how secure my thermostat would be if it were accessible online.	The thermostat controls are secure from unauthorized access.	The thermostat must be secure from unauthorized access.

Do not write the solution

Attributes help to formulate product specification

Do not lose the detail

Do not prioritize now!



#3: Organise the Needs into a Hierarchy

- Based on the customer survey analysis.
- Based on designer team knowledge, experience and intuition.
- Based on the competitor products

Key Points for Customer Need Analysis



- Express the need in terms of what the product has to do, not in terms of how it might do it.
- Express the need as specifically as the raw data.
- Use positive, not negative, phrasing.
- Express the need as an attribute of the product.
- Avoid the words must and should.

The *needs statements*, usually in hundreds, that are determined in the previous step are organised as:

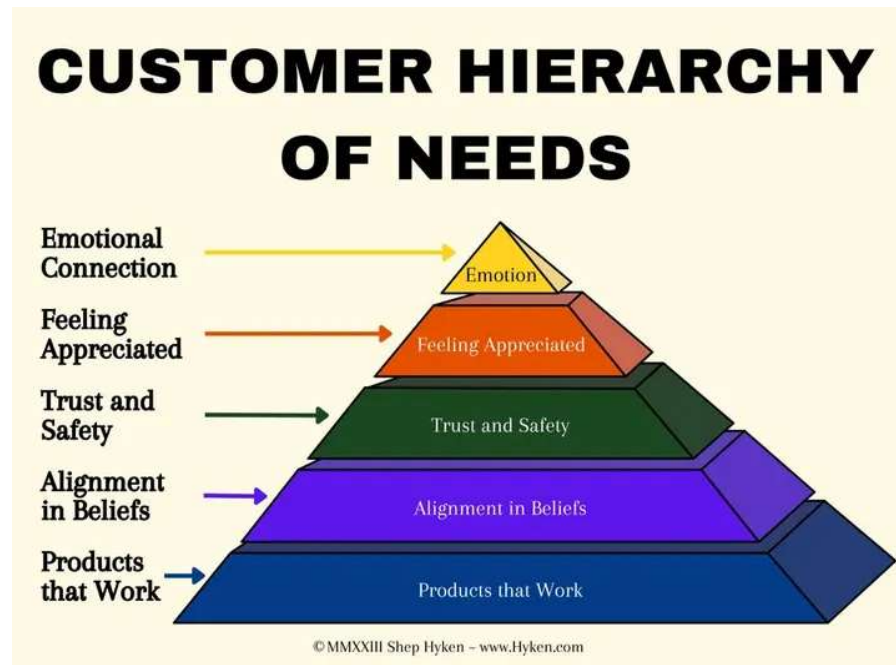
- (1) Primary Needs
- (2) Secondary Needs
- (3) Tertiary Needs (if required)

Step 3: Organize the Needs into a Hierarchy



- **Steps:**

1. Print or write each need statement on a separate card or self-stick note.
2. Eliminate redundant statements
3. Group the cards according to the similarity of the needs they express.
4. For each group, choose a label.
5. Consider creating supergroups consisting of two to five groups.
6. Review and edit the organized needs statements.



#3: Organising the Needs - Example



The thermostat controls are precise.

The thermostat maintains temperature accurately.
The thermostat minimizes unintended variability in temperature.
The thermostat allows temperatures to be specified precisely.

The thermostat is a good investment.

The thermostat is affordable to purchase.
The thermostat saves energy.
The thermostat tracks cost savings.

The thermostat is smart.

The thermostat can adjust temperature during the day according to user preferences.
The thermostat can be programmed to a precise schedule.
The thermostat automatically responds to occupancy.
The thermostat prevents pipes from freezing in cold months.
The thermostat alerts the user when a problem arises.
The thermostat does not require users to set time or date.
The thermostat adjusts automatically to the seasons.



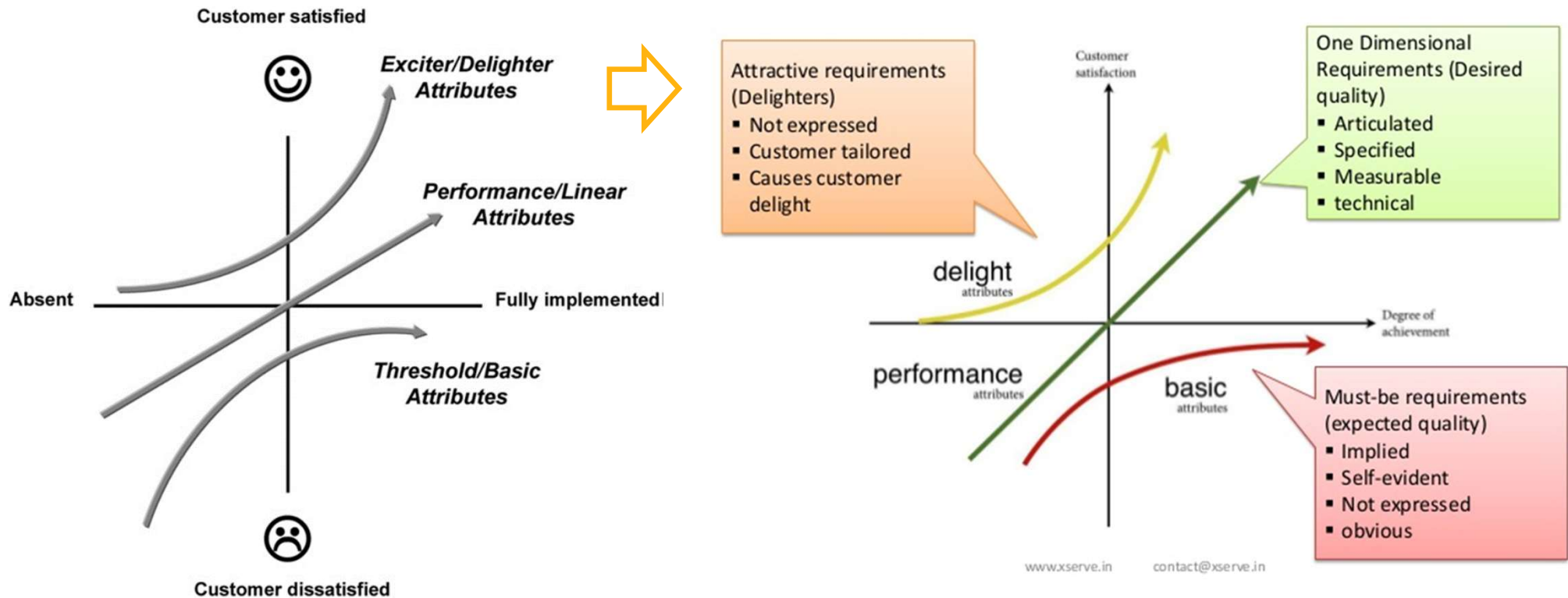
#4: Establish the Relative Importance of the Needs

Classifying Customer Requirements



- **Challenges in knowing the customers' voice:**
 - Customer must understand the product that is being designed.
 - Difficult situation in rapid technology change.
 - Typical problem is customer will always discuss about the failings of the product. Experience about the product usage will be never discuss easily
 - Kano recognized four level of customer requirements
 - (1) Expecters (2) Spoken (3) Unspoken (4) Exciters

Kano's Model – Alternative Approach



Eg. Top load washing machine



Kano Analysis



MUST BE	ONE DIMENSIONAL	DELIGHTER
Cabinet cost same as existing cabinet design	New Cabinet cost is \$5 cheaper compared to existing design	New Cabinet cost is \$10 cheaper compared to existing design
Cabinet must have same noise level as existing design	Noise level less than existing design	<50db noise level
Cabinet for WU capacity of 5.0cuft	Cabinet for WU capacity of 5.3cuft	Cabinet for WU capacity upto 5.5cuft

Kano's Model: Example



- Product requirements of skis



Product requirements
of skis

- Good edge grip on hard pistes
- Great ease of turn
- Good powder snow features
- Very light skis
- Integrated anti-theft device
- Scratch-resistant surface
- Design matches bindings and ski boots
- Free service of edges and base
- Trade-in offer for old skis
- Regular up-to-date information concerning test results, maintenance of ski and safety measures

Kano's Model



Functional form of the question If the edges of your skis grip well on hard snow, how do you feel? If the skis are light how do you feel ?	1. I like it that way 2. It must be that way 3. I am neutral 4. I can live with it that way 5. I dislike it that whay
Dysfunctional form of the question If the edges of your skis do not grip well on hard snow, how do you feel? If the skis are not light how do you feel ?	1. I like it that way 2. It must be that way 3. I am neutral 4. I can live with it that way 5. I dislike it that whay

Kano's Model



Customer Survey Responses		Dysfunctional Question Answer				
		1. Like	2. Must Be	3. Neutral	4. Live With	5. Dislike
Functional Question Answer	1. Like	<i>Questionable</i>	Attractive	Attractive	Attractive	One-Dimensional
	2. Must Be	Reverse	Indifferent	Indifferent	Indifferent	Must-Be
	3. Neutral	Reverse	Indifferent	Indifferent	Indifferent	Must-Be
	4.. Live With	Reverse	Indifferent	Indifferent	Indifferent	Must-Be
	5. Dislike	Reverse	Reverse	Reverse	Reverse	One-Dimensional

Kano's Model



- Indifferent category (I): Customer does not care whether it is present or not. He or she is, however, not willing to spend more on this feature.
- Questionable category (Q): Questionable scores signify that the question was phrased incorrectly, or that the person interviewed misunderstood the question or crossed out a wrong answer by mistake.
- Reverse Category (R): this product feature is not only not wanted by the customer but he even expects the reverse.

Kano Model



- Evaluation rule **M>O>A>I**
- Customer satisfaction coefficient (**CS coefficient**):

Product requirement	A	O	M	I	R	Q	Total	Category
Edge grip	7	32.3	49.3	9.5	0.3	1.5	100%	M
Ease of turn	10.4	45.1	30.5	11.5	1.2	1.2	100%	O
Service	63.8	21.6	2.9	8.5	0.7	2.5	100%	A



$$\text{Extent of Satisfaction} = \frac{A+O}{A+O+M+I}$$

$$\text{Extent of dissatisfaction} = \frac{O+M}{(A+O+M+I) \times (-1)}$$

Kano's Model



Product requirement	A	O	M	I	Total	Category	$\frac{A+O}{A+O+M+I}$	$\frac{O+M}{A+O+M+I}$
Edge grip	7	33	50	10	100%	M	0.40	-0.83
Ease of turn	11	46	31	12	100%	O	0.57	-0.78
Service	66	22	3	9	100%	A	0.89	-0.25

Advantages of Kano's Method



- Priorities customer requirements for product development.
- Product requirements are better understood.
- Kano's method provides valuable help in trade-off situations in the product development stage.
- Discovering and fulfilling attractive requirements creates a wide range of possibilities for differentiation.

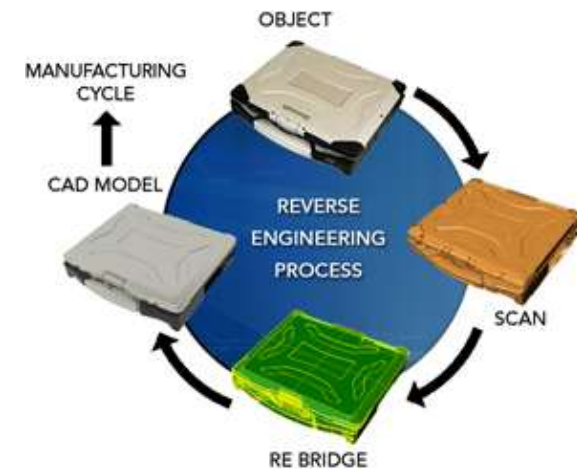
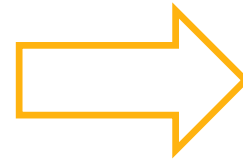
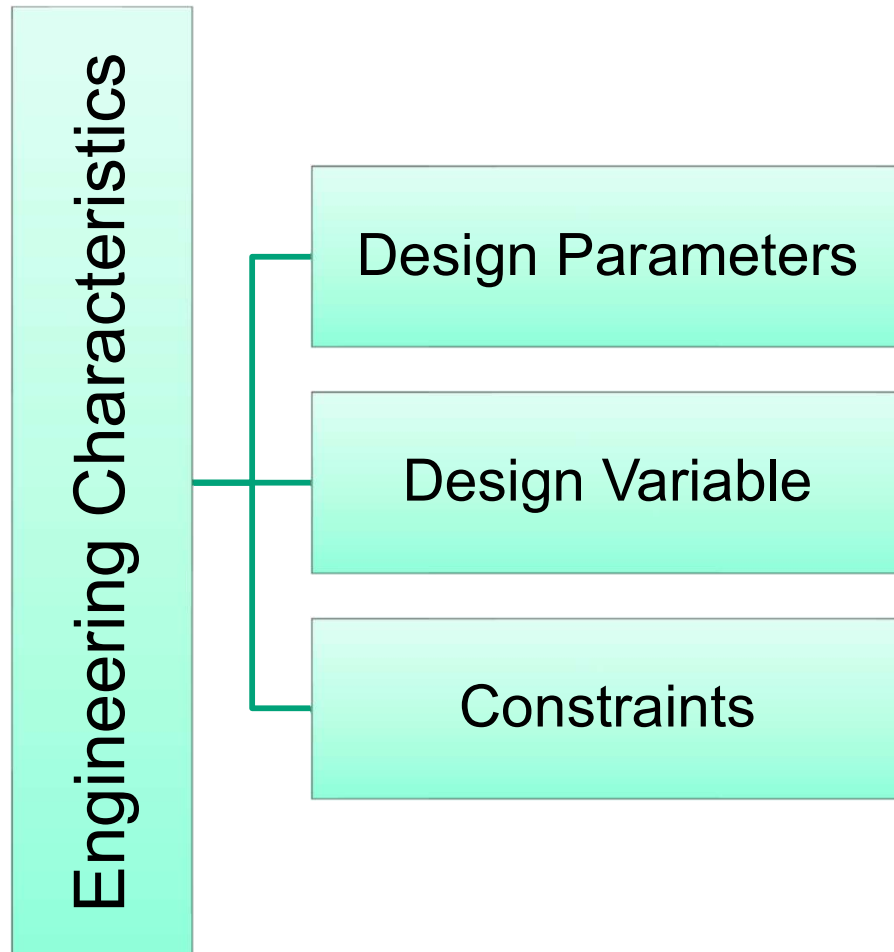
Disadvantages of Kano Model



“It’s really hard to design products by focus groups. A lot of times, people don’t know what they want until you show it to them.” - Steve Jobs.

- It can only be used for analysing the effects and certainly not for suggesting a new product feature, something that is quite difficult to achieve.
- It is based on more effective competition over quality.

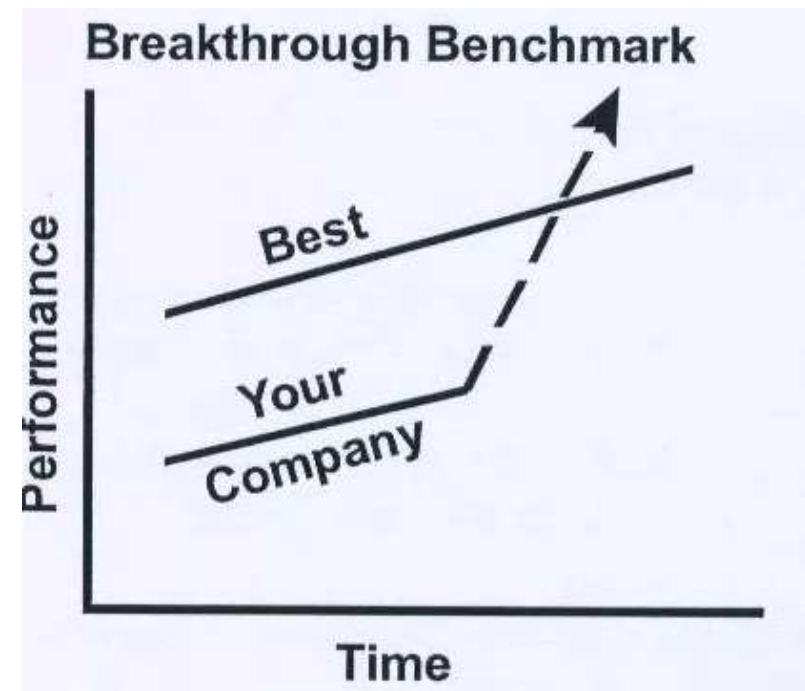
Establishing the Engineering Characteristics



Benchmarking



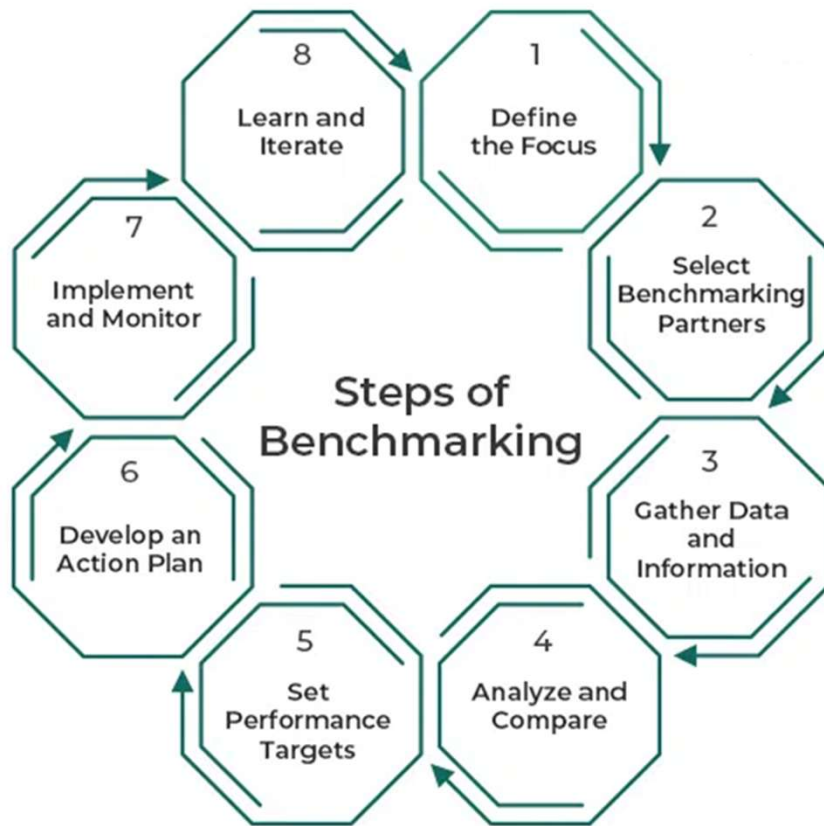
- Sun Tzu, Chinese warrior in 500 BC: “Know your enemy before to know yourself”
- Xerox’s worry of 1979 about Canon copier!
- **Benchmarking** is the key
- What is benchmarking?
- Why it is necessary?
- Identifying intra-company best practices



Importance of Competitive Benchmarking



Steps and Types of Benchmarking Process



Competitive / Product Benchmarking



PRODUCT(S) COVERED		TARGET CUSTOMER					
		COMPANY		COMPETITOR 1		COMPETITOR 2	
Value Proposition Dimensions	Customer Importance (Rank Order)	Core Features	Rating	Core Features	Rating	Core Features	Rating
Key Feature 1 	1	•XX	1	•XX	1	•XX	1
Key Feature 2	1	•XX	1	•XX	1	•XX	1
Aesthetics 	1	•XX	1	•XX	1	•XX	1
Usability 	1	•XX	1	•XX	1	•XX	1
Performance 	1	•XX	1	•XX	1	•XX	1
Durability / Reliability 	1	•XX	1	•XX	1	•XX	1
Serviceability 	1	•XX	1	•XX	1	•XX	1
Cost 	1	•XX	1	•XX	1	•XX	1
TOTALS			10		10		10

Resistance to Benchmarking Process



- The more common sources of resistance to benchmarking:
 - Fear of being perceived as copiers
 - Fear of yielding competitive advantages if information is traded or shared
 - Arrogance
 - Impatience

Pros and Cons of Benchmarking Process



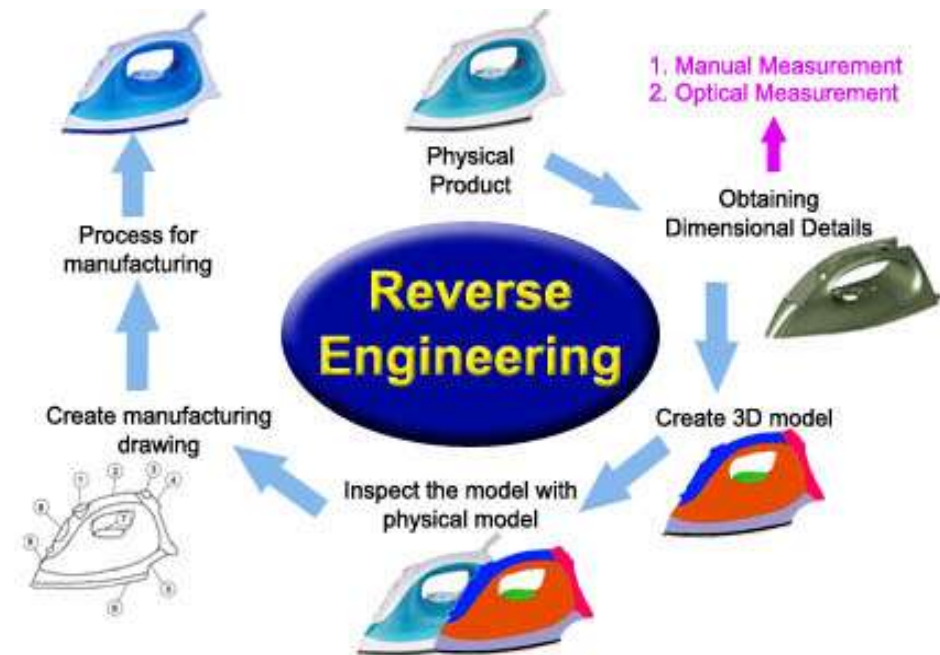
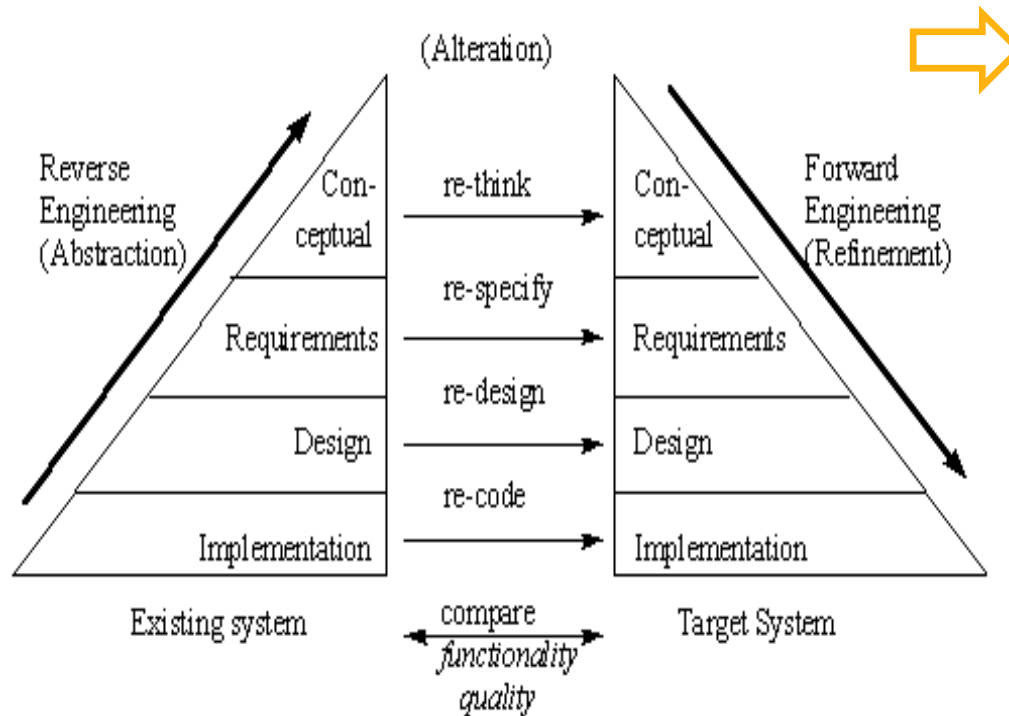
Advantages of Benchmarking



Disadvantages of Benchmarking



Reverse Engineering and Redesign



Reverse Engineering Process



1. Prediction:

- What is the purpose of this product?
- How does it work?
- What market was it designed to appeal to?
- List some of the design objectives for the product.
- List some of the constraints that may have influenced the design.

2. Observation:

- How do you think it works?
- How does it meet design objectives (overall)?
- Why is it designed the way it is?

3. Disassemble:

- How does it work?
- How is it made?
- How many parts?
- How many moving parts?
- Any surprises?

Reverse Engineering Process



4. Analyze:

- Carefully examine and analyze subsystems (i.e. structural, mechanical, and electrical).
- Develop annotated sketches that include measurements and notes on components, system design, safety, and controls.

5. Test:

- Carefully reassemble the product.
- Operate the device and record observations about its performance in terms of functionality (operational and ergonomic) and projected durability.

6. Documentation:

- Inferred design goals
- Inferred constraints
- Design (functionality, form (geometry), and materials)
- Schematic diagrams
- Lists (materials, components, critical components, flaws, successes, etc.)
- Identify any refinements that might enhance the product's usefulness.
- Upgrades and changes

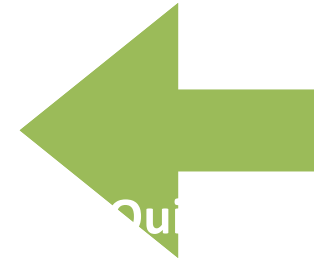
#4: Importance of the Needs



It is hard to satisfy all the needs in the product, and hence, it is natural to determine the importance of the needs through trade-offs.

This can be done by:

- (1) Consensus of the team members (development team)
- (2) Further customer surveys





#5: Reflect on the Results and the Process

#5: Things to be Reflected Upon



It is important to challenge the results of step #1 to #4, by reviewing them thoroughly. Following aspects would help:

- (1) Important customers
- (2) Latent needs identification
- (3) Pending inquiries or follow-ups
- (4) Customers to involve during PD efforts
- (5) Any 'surprising' needs
- (6) Involvement of relevant stakeholders
- (7) Any improvement in the process

Summarise the results by listing Most Important Needs and Latent Needs!



Summary

Summary



- (1) Identifying customer needs is an integral part of the concept development phase of the product development process
- (2) The process of identifying customer needs includes five steps:
 - *Gather raw data from customers*
 - *Interpret the raw data in terms of customer needs*
 - *Organise the needs into a hierarchy*
 - *Establish the relative importance of the needs*
 - *Reflect on the results and the process*
- (3) It is crucial to have a high-quality information channel from customers to the product developers

Summary



- (4) Lead users are a good source of customer needs, whereas Extreme users have special needs that may reflect latent needs
- (5) Expressing customer needs as “what product has to do” would provide maximum flexibility to the PD team
- (6) The key benefits of the method are ensuring that the product is focused on customer needs



Thank You!



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Product Design

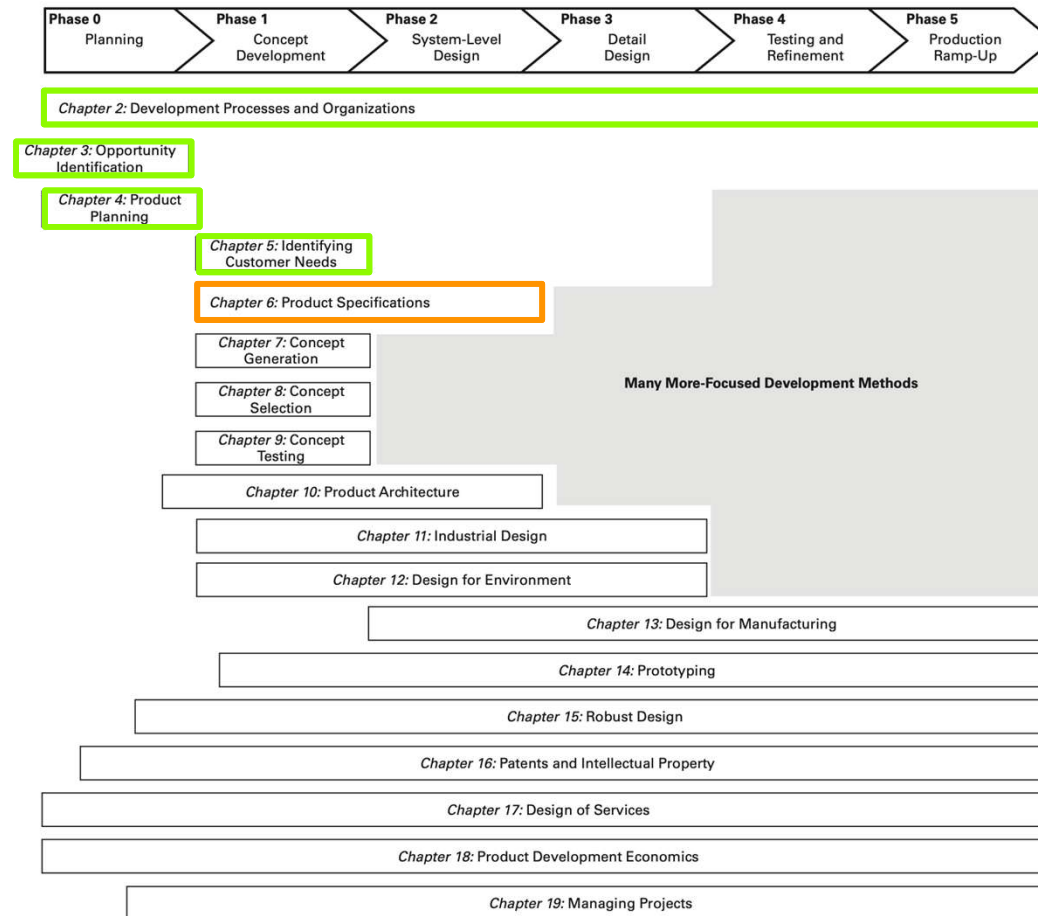
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Lecture-4: Develop Product Specifications

Course Summary





Topics to be covered:

- What are Product Specifications
- Steps to Establish “Target” Specifications
- Steps to Set “Final” Specifications
- Quality Function Deployment



What is Product Specification

Customer Need



Customer Statement	Interpreted Need
I have to manually turn it on and off when it gets too hot or cold.	The thermostat maintains a comfortable temperature without requiring user action.
Each time I want to change the temperature, I need to adjust both thermostats in the house.	Any user inputs need not be made in multiple locations. (!)
I like that I can change the temperature if the setting is too high.	The temperature setting is easy to control manually.
It didn't cost a fortune.	The thermostat is affordable to purchase.
I'm too lazy to learn how to figure it out.	The thermostat requires little or no user instruction or learning.
I sometimes forget to turn it off when we leave the house.	The thermostat saves energy when no one is home. (!)
Sometimes the buttons don't register a push and I have to press it repeatedly.	The thermostat definitively registers any user inputs.
I would like my iPhone to adjust my thermostat.	The thermostat can be controlled remotely without requiring a special device.
I like to shift quickly between different options like energy saving or ultra comfort.	The thermostat responds immediately to differences in occupant preferences.

What next?

- Translate to Precise Targets
- Metrics for Success or Failure
- Gaining Confidence
- Resolving Inevitable Trade-offs

Product Specifications



Customer Need

Language of the customer

Subjective

Specification

What the product has to do, ~~and how~~

Unambiguous

a metric and a value

Product Specification

Precise description of what product has to do

**Product Specification / Requirement / Technical Specification or
Engineering Characteristics**

Technical Specification of a Car



Technical Description		1.0 TSI/85 kW		1.5 TSI/110 kW
Engine		MT	AT	DSG
Engine type		1.0 TSI Petrol , in-line, 4 Stroke Positive Ignition Direct Injection Engine		1.5 TSI Petrol , in-line, 4 Stroke Positive Ignition Direct Injection Engine with Active Cylinder Technology
Clutch		Dry Single Plate	Hydro Dynamic Torque Converter	Dry Double Clutch
Cylinders		3		4
Displacement	[cm³]	999		1498
Max. power/revs	[kWat rpm]	85/ 5000-5500		110 / 5000-6000
Max. torque/revs	[Nm at rpm]	178 / 1750-4500		250 / 1600-3500
Fuel consumption - rule 115 of CMVR	[kmp/l]	20.32	18.73	19.36
Transmission				
Wheel drive		Front Wheel Drive		
Transmission		6 Speed	6 Speed	7 Speed
Chassis				
Front axle		McPherson suspension with lower triangular links and stabiliser bar		
Rear axle		Twist Beam Axle		
Braking system		Hydraulic Diagonal Split Vaccum Assisted Braking System		
Brake – front		Disc		
Brake – rear		Drum		
Steering system		Electro-Mechanical Steering (Power Assisted)		
Wheels		Steel wheel 6J x 15" (Classic), Alloy wheel 6J x 16" (Signature/Sportline/Prestige/Monte Carlo)		
Tyres		195/65 R15-Classic, 205/55 R16-Signature/Sportline/Prestige/Monte Carlo		
Outside dimensions				
Length	[mm]	4 541		
Width	[mm]	1 752		
Height	[mm]	1 507		
Wheel base	[mm]	2 651		
Ground clearance (Laden condition)	[mm]	145		

www.skoda-auto.co.in

Example: Customer Need & Specification



Customer Need

Make flights cheaper

Specification

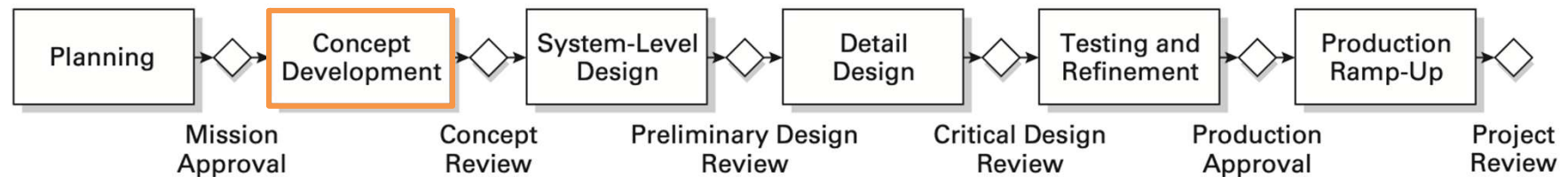
Specific Fuel Consumption is
less than 0.207 Kg/hr/kW

a *metric* and a *value*

When Does it Happen?

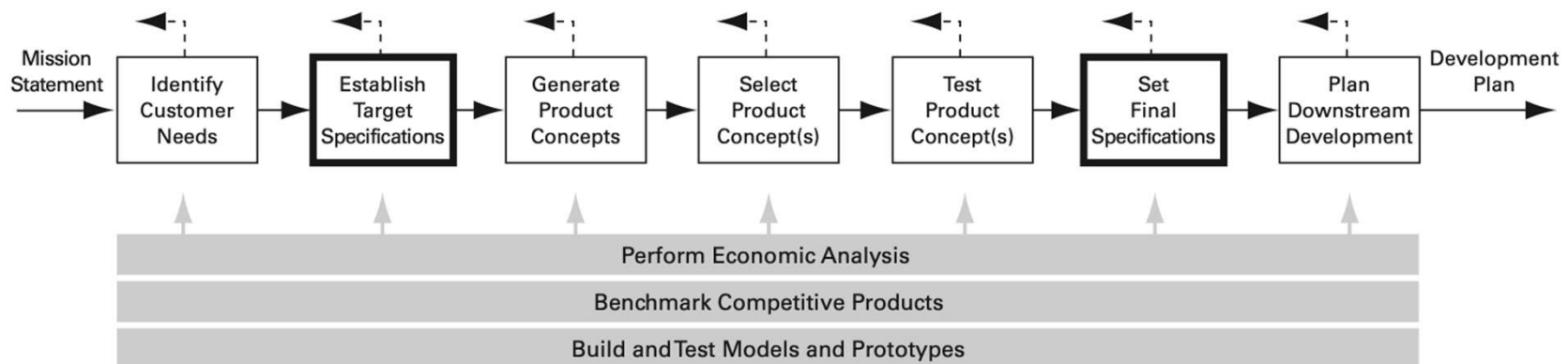


Generic PD Process



Concept Development

Get to know more about constraints, cost, etc.





Establish Target Specifications

What is Target Specification



Target specifications are “preliminary” specifications that are set as “goals” by the PD team.

It cannot be “arbitrary”, as it may not be technically feasible.

E.g. 1400 Kg of mass + manufacturing cost of 10 L + power of 100 kW

Steps in Establishing Target Specifications



The Target specifications can be established in four steps:

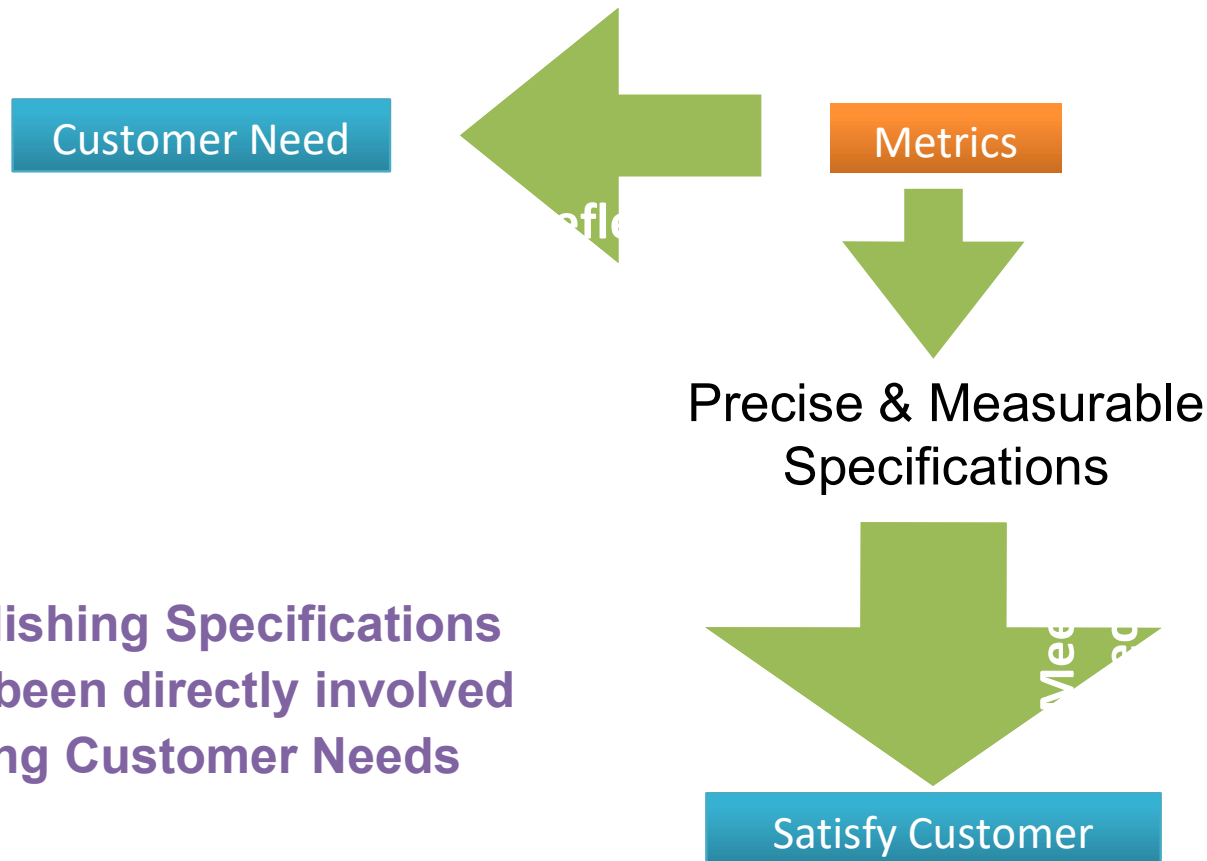
- (1) Prepare the **list of metrics**
- (2) Collect competitive **benchmarking** information
- (3) Set **ideal** and **marginally acceptable** target values
- (4) **Reflect** of the result and the process



Step 1: Prepare the List of Metrics

Establish Target Specifications

#1: Metrics Vs. Customer Need



Those establishing Specifications should have been directly involved in identifying Customer Needs

#1: List of Metrics



Metric No.	Need Nos.	Metric
1	1, 3	Attenuation from dropout to handlebar at 10 Hz
2	2, 6	Spring preload
3	1, 3	Maximum value from the Monster
4	1, 3	Minimum descent time on test track
5	4	Damping coefficient adjustment range
6	5	Maximum travel (26-in. wheel)
7	5	Rake offset
8	6	Lateral stiffness at the tip
9	7	Total mass

#1: Needs-Metrics Matrix



For some cases, matrix creation/mapping can be complex!

		Metric								
		1	2	3	4	5	6	7	8	9
		Attenuation from dropout to handlebar at 10 Hz								
		Spring preload								
		Maximum value from the Monster								
		Minimum descent time on test track								
		Damping coefficient adjustment range								
		Maximum travel (26-in. wheel)								
		Rake offset								
		Lateral stiffness at the tip								
		Total mass								
1	Reduces vibration to the hands	●		●	●					
2	Allows easy traversal of slow, difficult terrain		●							
3	Enables high-speed descents on bumpy trails	●		●	●					
4	Allows sensitivity adjustment					●				
5	Preserves the steering characteristics of the bike						●	●		
6	Remains rigid during hard cornering		●						●	
7	Is lightweight									●

#1: Guidelines to Construct Metrics



The metrics should be

Complete

Correlate Perfectly

Dependent

What-not-how

Freedom to Designers

E.g. Fuel consumption

Practical

Easily measurable

Cylinders		3
Displacement	[cm ³]	999
Max. power/revs	[kW at rpm]	85/ 5000-5500
Max. torque/revs	[Nm at rpm]	178 / 1750-4500
Fuel consumption - rule 115 of CMVR	[kmpl]	20.32

#1: Guidelines to Construct Metrics



Also, the metrics

- (1) can be Subjective
E.g. Feels premium, instills pride
- (2) should include criterias from independently published evaluations
E.g. ICOTY
The Indian Car of the Year Award

#1: Units and Importance of Metrics



Metric No.	Need Nos.	Metric	Imp.	Units
1	1, 3	Attenuation from dropout to handlebar at 10 Hz	3	dB
2	2, 6	Spring preload	3	N
3	1, 3	Maximum value from the Monster	5	g
4	1, 3	Minimum descent time on test track	5	s
16	11	Fender compatibility	1	List
17	12	Instills pride	5	Subj.
25	20	Japan Industrial Standards test	5	Binary
26	20	Bending strength (frontal loading)	5	kN

Engineering units

List

Subjective

Pass or Fail

Importance rating of Metrics are usually derived from the importance ratings of the needs

#1: IBM Customer Service Metrics



Customer satisfaction score (CSAT)

Customer effort score (CES)

Net promoter score (NPS)

Social media metrics

Customer churn

First response time

First response time, or FRT, measures the amount of time it takes for an agent to provide an initial reply to a customer's inquiry or complaint. This metric is crucial because quicker responses to customer queries often lead to a better customer service experience and the perception of better service quality. By monitoring FRT, businesses can identify areas for improvement in their response processes and help ensure that customers feel valued and heard promptly.

How to measure: Businesses track the timestamp of when a customer inquiry is received and the timestamp of when the first response is sent. The average time taken across multiple inquiries is then calculated, providing a clear metric that can be analyzed over time to assess performance and identify trends.

Average ticket resolution rate

Average handle time

Preferred communication channel

Customer retention rate

First contact resolution rate (FCR)

Service level agreement (SLA) rate

<https://www.ibm.com/think/insights/top-customer-service-metrics-you-should-be-measuring>



Step 2: Collect Competitive Benchmarking Information

Establish Target Specifications

#2: Need of Benchmarking



Relationship of the New product
with Competitive products



“Target Specifications” is the “Language” for
the PD team to achieve commercial success!

www.skoda-auto.co.in

<https://www.hondacarindia.com/honda-city-5th-generation>

<https://www.hyundai.com/in/en/find-a-car/verna/highlights>

#2: Benchmarking Chart-1



In terms of “Metrics”

Metric No.	Need Nos.	Metric	Imp.	Units	ST Tritrack	Maniray 2	Rox Tahx Quadra	Rox Tahx Ti 21
1	1, 3	Attenuation from dropout to handlebar at 10 Hz	3	dB	8	15	10	15
2	2, 6	Spring preload	3	N	550	760	500	710
3	1, 3	Maximum value from the Monster	5	g	3.6	3.2	3.7	3.3
4	1, 3	Minimum descent time on test track	5	s	13	11.3	12.6	11.2
5	4	Damping coefficient adjustment range	3	N-s/m	0	0	0	200
6	5	Maximum travel (26-in. wheel)	3	mm	28	48	43	46
7	5	Rake offset	3	mm	41.5	39	38	38
8	6	Lateral stiffness at the tip	3	kN/m	59	110	85	85
9	7	Total mass	4	kg	1.409	1.385	1.409	1.364

Often expensive and time consuming.....
(Purchasing, testing, dismantling, cost estimation, investigation, etc.)

#2: Benchmarking Chart-2



In terms of “Customer Needs”.... “relative” based on Customer’s perception

No.	Need	Imp.	ST Tritrack	Maniray 2	Rox Tahx Quadra	Rox Tahx Ti 21
1	Reduces vibration to the hands	3	•	••••	••	•••••
2	Allows easy traversal of slow, difficult terrain	2	••	••••	•••	•••••
3	Enables high-speed descents on bumpy trails	5	•	•••••	••	•••••
4	Allows sensitivity adjustment	3	•	••••	••	•••••
5	Preserves the steering characteristics of the bike	4	••••	••	•	••
6	Remains rigid during hard cornering	4	•	•••	•	•••••
7	Is lightweight	4	•	•••	•	•••

Can be expensive and time consuming

#2: Example of Benchmarking Chart



☐ Hide common features
☐ Highlight differences

	Tata Nexon Smart Petrol 1.2... Get Offers	Hyundai Venue HX 2 Petrol 1.2L... Get Offers	Skoda Kylaq Classic Petrol... Get Offers	Toyota Urban... E 2WD Petrol... Get Offers
--	--	---	---	---

Specifications Features Brochure Colours User Reviews

Search a spec o

Ground Clearance (Know More)	208 mm (unladen)	190 mm (unladen)	189 mm (unladen)	—
Wheelbase (mm) (Know More)	2498	2520	2566	2600
Kerb Weight Range (Know More)	—	—	1169 - 1219 kg	1145 - 1190 kg
Gross Weight (kg) (Know More)	—	—	1630	1650
Capacity				
Seating Capacity (Know More)	5 Seats and 2 Rows	5 Seats and 2 Rows	5 Seats and 2 Rows	5 Seats and 2 Rows
Doors (Doors) (Know More)	5	5	5	5
Bootspace (Know More)	382 L	375 L	360 L (measured up to the roof) – 1265 L (rear seats folded)	373 L
Fuel Tank Capacity (litres) (Know More)	44	45	45	45

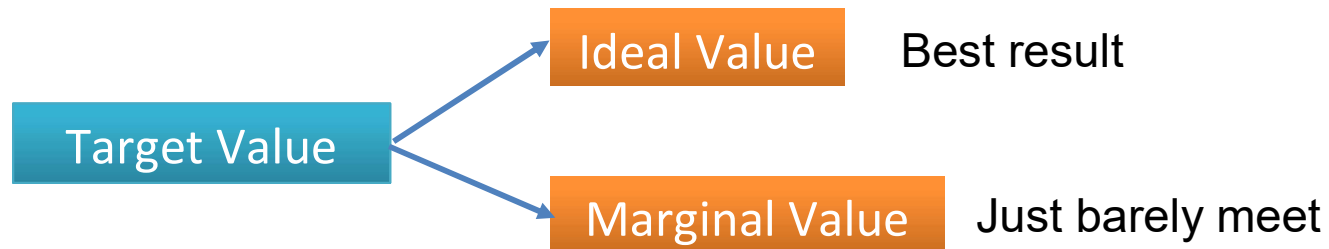
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Step 3: Set Ideal and Marginally Acceptable Target Values

Establish Target Specifications

#3: Target Value



#3: Example of Target Values



Metric No.	Need Nos.	Metric	Imp.	Units	Marginal Value	Ideal Value
1	1, 3	Attenuation from dropout to handlebar at 10 Hz	3	dB	>10	>15
2	2, 6	Spring preload	3	N	480–800	650–700
3	1, 3	Maximum value from the Monster	5	g	<3.5	<3.2
4	1, 3	Minimum descent time on test track	5	s	<13.0	<11.0
5	4	Damping coefficient adjustment range	3	N-s/m	0	>200
6	5	Maximum travel (26-in. wheel)	3	mm	33–50	45
7	5	Rake offset	3	mm	37–45	38
8	6	Lateral stiffness at the tip	3	kN/m	>65	>130
9	7	Total mass	4	kg	<1.4	<1.1

The values can be
exact, discrete, at least, at most, between



Step 4: Reflect on the Results and the Process

Establish Target Specifications

#4: Reflection



- (1) Is the team influenced by 'marketing' and 'sales'?
e.g. specifications driven by marketing and sales assumptions rather than objective user research
- (2) Single Vs. multiple products - To match needs of various markets
- (3) Missing specifications
- (4) Dictating commercial success